

ARTIFICIAL INTELLIGENCE

Lecture Set 09

Knowledge Representation and Logical Agents

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Knowledge Representation

- It is responsible for representing information about the real world so that a computer can understand and can utilize this knowledge to solve the complex real world problems such as diagnosis a medical condition or communicating with humans in natural language.
- It is also a way which describes how we can represent knowledge in artificial intelligence. Knowledge representation is not just storing data into some database, but it also enables an intelligent machine to learn from that knowledge and experiences so that it can behave intelligently like a human.

Propositional Logic

- Proposition means sentences.
- Propositional logic applies the Boolean logic to **convert** our **real-world data** into a format that is **readable to the computer**.
- For instance, if we say 'It is hot and humid today', the machine won't understand. But if we can create propositional logic for this sentence, then, we can make the machine-read, and interpret our message.
- Derived from Boolean logic, the heart of propositional logic is the idea that the final output (meaning) of all propositions are either true or false. It can't be both.
- For example, '**Earth is round**', the output for this proposition is **TRUE**. If we say, '**Earth is square**', then the output is **FALSE**.
- **Propositional logic** applies to those sentences where the output can only be either **TRUE or FALSE**.

Propositional Logic

- But if we refer to the sentence like 'Some children are lazy' then here we have two possible outputs.
- This proposition is TRUE for those children who are lazy, but it is FALSE for those children who are not lazy.
- So, for such sentences/propositions where two or more outputs are possible, propositional logic doesn't apply.

Types of Propositions

- There are two types of propositions, atomic and complex, that can be represented through propositional logic.
- **Atomic** means a single proposition like '**the sky is blue**', 'hot days are humid', water is liquid, etc.
- **Complex propositions** are those, which have been formed by **connecting one, two, or more sentences**.

Symbols

- In propositional logic, there are five symbols to create the syntax to represent the connection of two or more sentences. Syntax means a proper structure to represent information.

	<i>Symbol</i>	<i>Meaning</i>
<i>Negation</i>	$\sim$	<i>NOT</i>
<i>Conjunction</i>	$\wedge$	<i>AND</i>
<i>Disjunction</i>	$\vee$	<i>OR</i>
<i>Conditional</i>	$\rightarrow$	<i>If/Then</i>
<i>Biconditional</i>	$\longleftrightarrow$	<i>IF and ONLY IF</i>

Example

- **Ram can play tennis** (let's take it as variable **X**)
- **Ram cannot play tennis** – There is a negation in the sentence, so symbolic representation will be $\sim X$
- **Ram can play tennis and badminton** – Note, there is a new addition 'Badminton', let's take it as variable **Y**. Now, this sentence has a Conjunction, so symbolic representation will be $X \wedge Y$
- **Ram can play tennis or badminton** – Here is a Disjunction, so symbolic representation will be $X \vee Y$
- **If Ram can play tennis then he can play badminton** – There is a condition, so symbolic representation will be $X \rightarrow Y$
- **Ram can play tennis if and only if he can play badminton** – It is a biconditional sentence, so symbolic representation will be $X \leftrightarrow Y$

Disadvantages of PL

- If you want to represent complicated sentences or natural language statements, PL is not sufficient.
- There is very limited expressive power in PL, so we use FOL instead.
- The sentence shown below cannot be represented using PL logic.

Examples

- I love mankind. It's the people I can't stand!
- Joe Root likes football.
- I like to eat mangos.

PL is not enough to represent the sentences above, so we require powerful logic (such as FOL).

Predicate Logic or First-Order Logic (FOL)

- FOL is also called *predicate logic*. It is a powerful language used to develop information about an object and express the relationship between objects.
- FOL not only assumes that does the world contains facts (like PL does), but it also assumes the following:
 - **Objects:** A, B, people, numbers, colors, wars, theories, squares, pit, etc.
 - **Relations:** It is unary relation such as red, round, sister of, brother of, etc.
 - **Function:** father of, best friend, third inning of, end of, etc.

Parts of First-Order Logic

FOL also has two parts:

- Syntax
- Semantics

Syntax

- The **syntax** of FOL decides which collection of symbols is a logical expression.
- The basic syntactic elements of FOL are symbols. We use symbols to write statements in shorthand notation.

Basic Elements of FOL

Name	Symbol
Constant	1, 6, A, W, New York, Elie, Dog...
Variables	a, b, c, x, y, z...
Predicates	<, >, brother, sister, father...
Equality	==
Function	Sqrt, LessThan, Sin(θ)...
Quantifier	$\forall$, $\exists$
Connectives	$\wedge$, $\vee$, $\neg$, $\Rightarrow$, $\Leftrightarrow$

Atomic and Complex Sentences in FOL

1. Atomic Sentence

- This is a basic sentence of FOL formed from a predicate symbol followed by a parenthesis with a sequence of terms.
- We can represent atomic sentences as a predicate (value1, value2..., value n).

Example

1. John and Michael are colleagues → **Colleagues (John, Michael)**
2. German Shepherd is a dog → **Dog (German Shepherd)**

2. Complex sentence

Complex sentences are made by combining atomic sentences using connectives.

FOL is further divided into two parts:

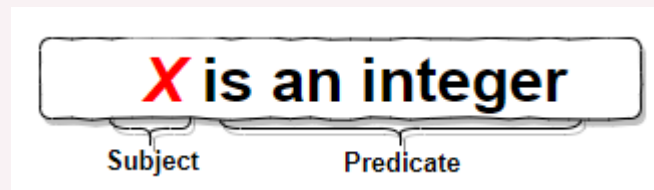
- **Subject:** the main part of the statement.
- **Predicate:** defined as a relation that binds two atoms together.

Example

1. Colleague (Oliver, Benjamin) **∧** Colleague (Benjamin, Oliver)
2. "x is an integer"

It has two parts;

- first, **x** is the subject.
- second, **"is an integer"** is called a predicate.



Quantifiers and their use in FOL

- Quantifiers allow us to determine or identify the range and scope of the variable in a logical expression.
- There are two types of quantifiers:
 - **Universal quantifier**: for all, everyone, everything.
 - **Existential quantifier**: for some, at least one.

Universal Quantifiers

- **Universal quantifiers** specify that the statement within the range is **true** for everything or every instance of a particular thing.
- Universal quantifiers are denoted by a symbol ($\forall$) that looks like an inverted A. In a universal quantifier, we use $\rightarrow$.
- If x is a variable, then $\forall x$ can read as:
 1. For all x
 2. For every x
 3. For each x

Example

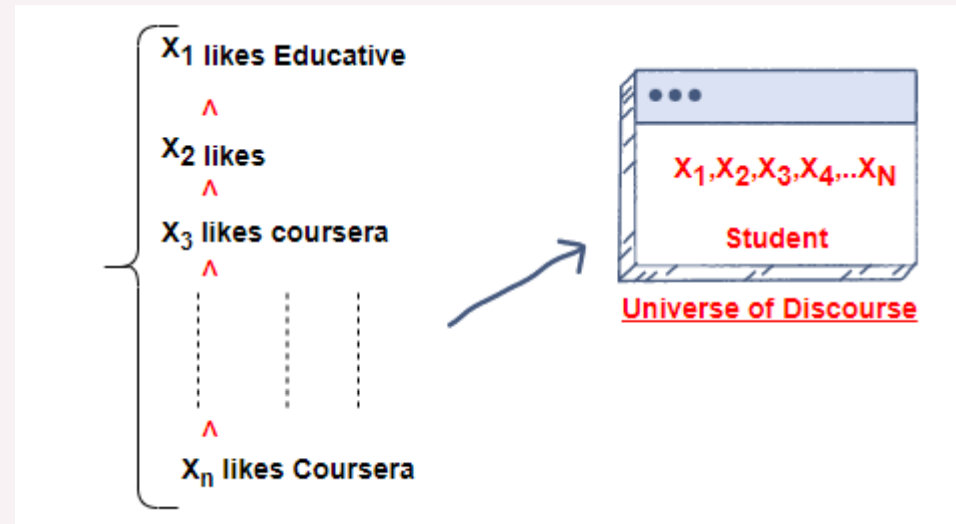
Every Student Likes Educative.

Explanation

So, in logical notation, it can be written as:

$\forall x \text{ student}(x) \rightarrow \text{likes}(x, \text{Educative})$

This can be interpreted as: *There is every x where x is a student who likes Educative.*



Existential Quantifiers

- **Existential quantifiers** are used to express that the statement within their scope is true for at least one instance of something.
- $\exists$, which looks like an inverted E, is used to represent them. We always use **AND** or conjunction symbols.
- If x is a variable, the existential quantifier will be $\exists x$:
 1. For some x
 2. There exists an x
 3. For at least one x

Example

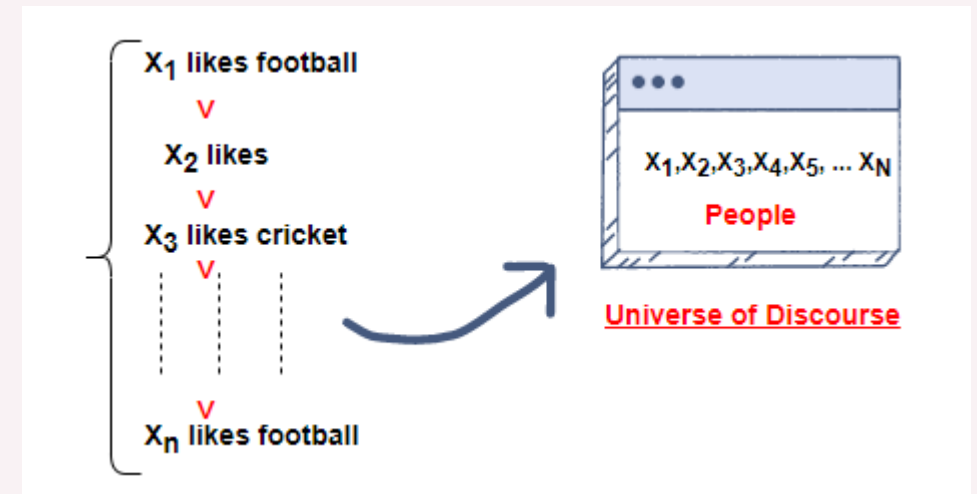
Some people like Football.

Explanation

So, in logical notation, it can be written as:

$\exists x: \text{people}(x) \wedge \text{likes Football}(x)$

It can be interpreted as: *There are some x where x is people who like football.*



Nested Quantifiers and their uses

We can use both quantifiers together, but it's not a type of quantifier; rather, it's an outlier category.

- **Nested quantifier** refers to when one quantifier is within the scope of another quantifier.
- These quantifiers can be represented using the $\exists x \forall x$ signs.
- Here are some examples to understand this type of quantifier.

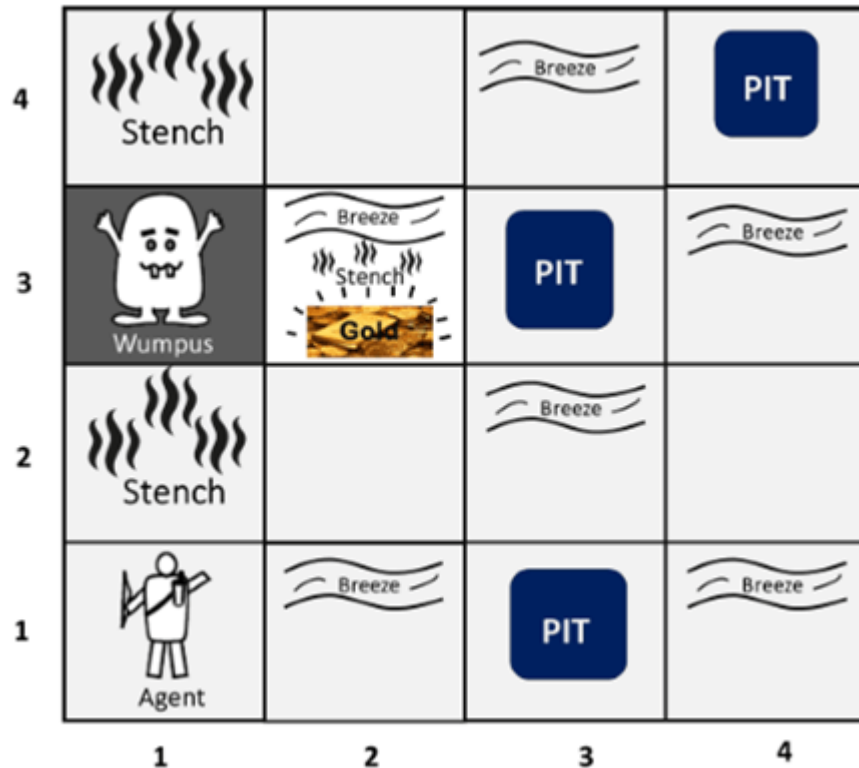
$$1. \exists xy \forall x \forall y ((x < 0) \wedge (y < 0) \rightarrow (xy = 8))$$

This can be interpreted as: *For every real number x and $y \in \mathbb{R}$, if x is negative and y is also negative, implies for some values of xy must be equal to 8.*

Sentences to Well formed Formulas in Predicate Logic

- | | |
|---|---|
| 1. Marcus was a man. | 1. $man(Marcus)$ |
| 2. Marcus was a Pompeian. | 2. $Pompeian(Marcus)$ |
| 3. All Pompeian were Romans. | 3. $\forall x : Pompeian(x) \rightarrow Roman(x)$ |
| 4. Caesar was a ruler. | 4. $ruler(Caesar)$ |
| 5. All Romans were either loyal to Caesar or hated him. | 5. $\forall x : Roman(x) \rightarrow loyalto(x, Caesar) \vee hate(x, Caesar)$ |
| 6. Everyone is loyal to someone. | 6. $\forall x : \exists y : loyalto(x, y)$ |
| 7. People only try to assassinate rulers they are not loyal to. | 7. $\forall x : \forall y : person(x) \wedge ruler(y) \wedge tryassassinate(x, y) \rightarrow \neg loyalto(x, y)$ |
| 8. Marcus tried to assassinate Caesar. | 8. $tryassassinate(Marcus, Caesar)$ |
| 9. All men are people | 9. $\forall x : man(x) \rightarrow person(x)$ |

Wumpus World (Propositional Logic Example)



There are also some components which can help the agent to navigate the cave.

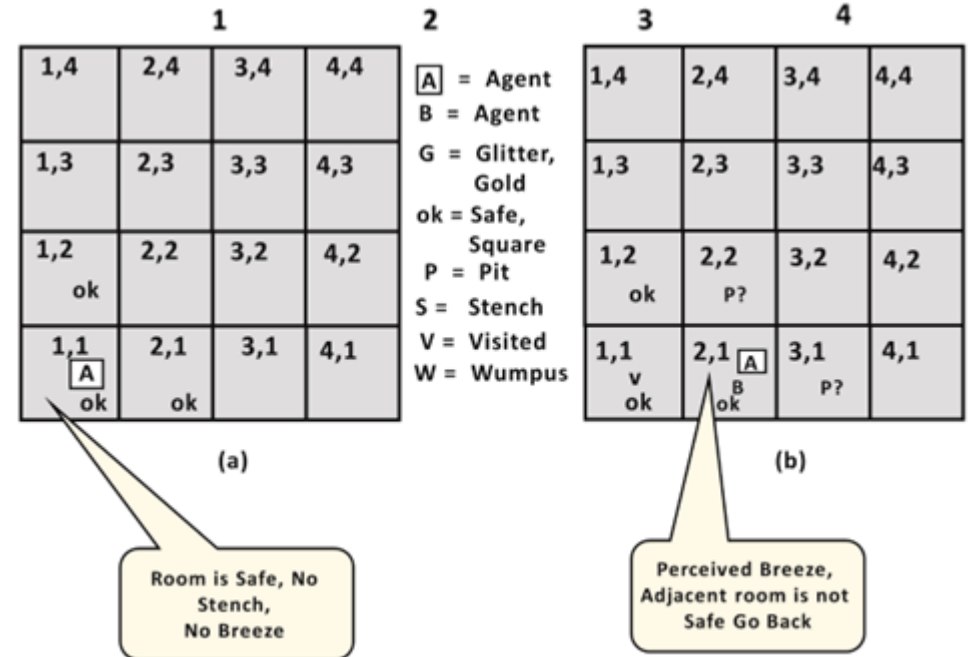
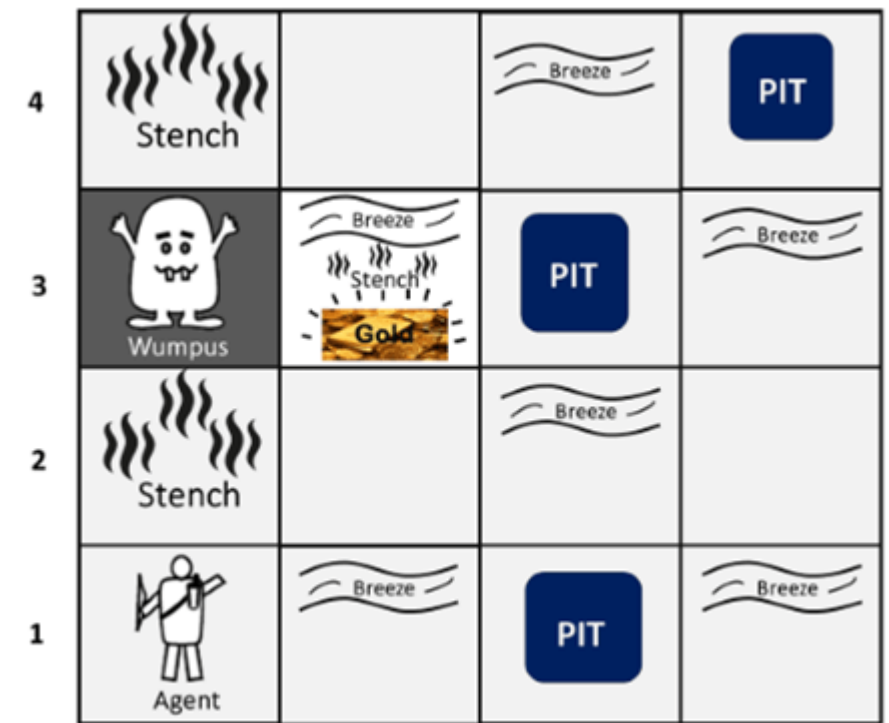
- 1.The rooms adjacent to the Wumpus room are smelly, so that it would have some stench.
- 2.The room adjacent to PITs has a breeze, so if the agent reaches near to PIT, then he will perceive the breeze.
- 3.There will be glitter in the room if and only if the room has gold.
- 4.The Wumpus can be killed by the agent if the agent is facing to it, and Wumpus will emit a horrible scream which can be heard anywhere in the cave.

- Left turn,
- Right turn
- Move forward, backward
- Grab

- **Agent's First & Second step:**

- Initially, the agent is in the first room or on the square [1,1], and we already know that this room is safe for the agent, so to represent on the below diagram (a) that room is safe we will add symbol OK. Symbol A is used to represent agent, symbol B for the breeze, G for Glitter or gold, V for the visited room, P for pits, W for Wumpus.

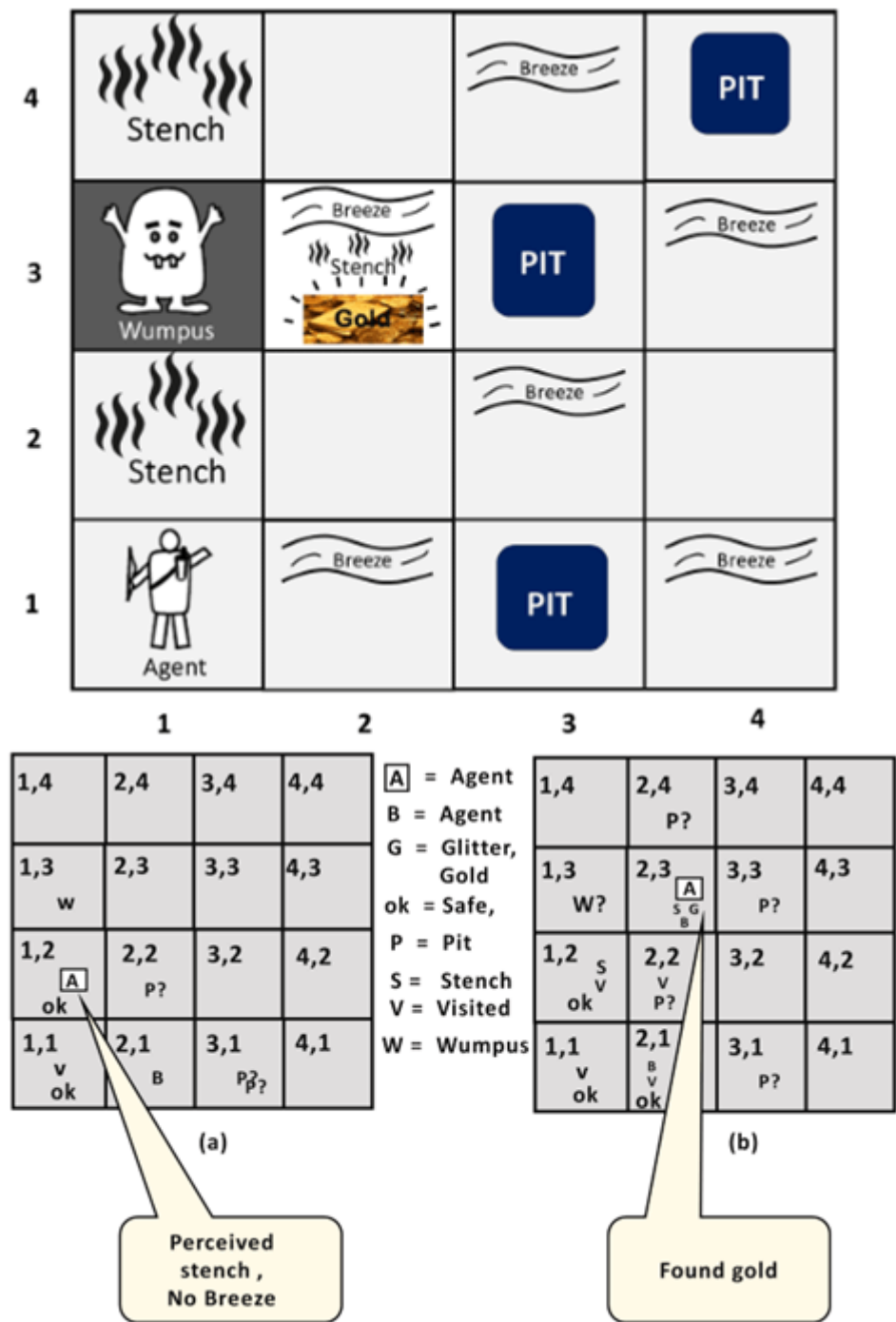
- At Room [1,1] agent does not feel any breeze or any Stench which means the adjacent squares are also OK.
- Now agent will stop and think and will not make any harmful move. The agent will go back to the [1, 1] room. The room [1,1], and [2,1] are visited by the agent, so we will use symbol V to represent the visited squares.



- **Agent's 3 & 4 Step:**

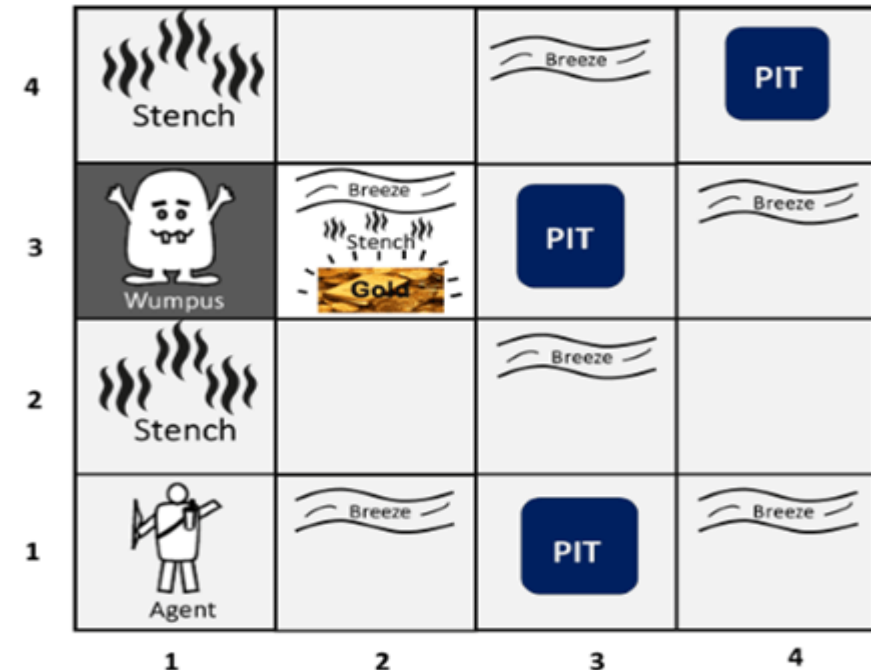
- But Wumpus cannot be in the room [1,1] as by rules of the game, and also not in [2,2] (Agent had not detected any stench when he was at [2,1]). Therefore agent infers that Wumpus is in the room [1,3], and in current state, there is no breeze which means in [2,2] there is no Pit and no Wumpus. So it is safe, and we will mark it OK, and the agent moves further in [2,2].

- At room [2,2], here no stench and no breezes present so let's suppose agent decides to move to [2,3]. At room [2,3] agent perceives glitter, so it should grab the gold and climb out of the cave.

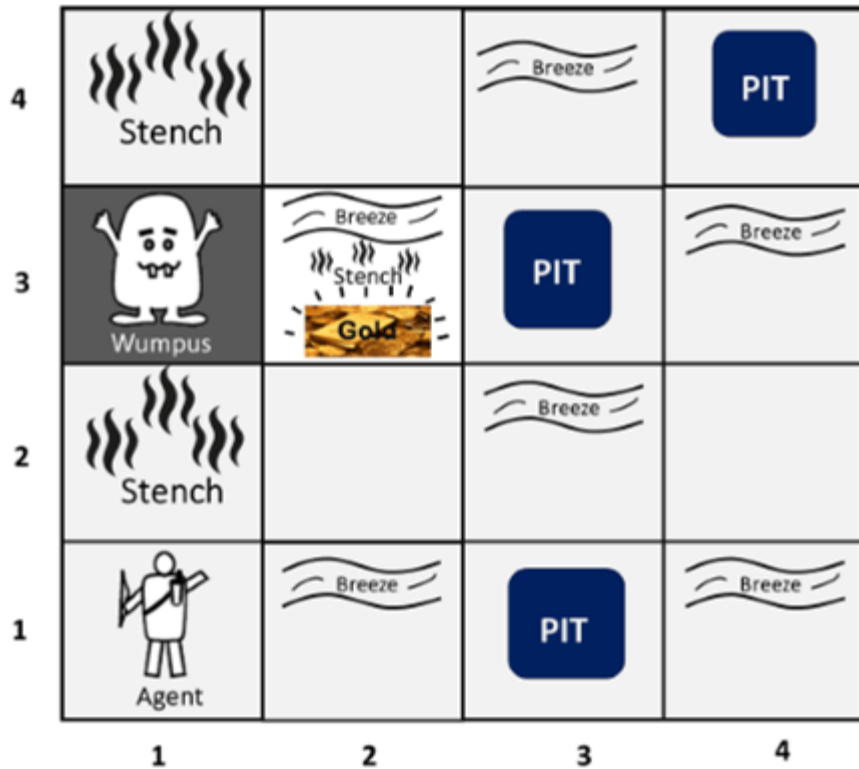


Atomic proposition variable for Wumpus world:

- Let $P_{i,j}$ be true if there is a Pit in the room $[i, j]$.
- Let $B_{i,j}$ be true if agent perceives breeze in $[i, j]$, (dead or alive).
- Let $W_{i,j}$ be true if there is wumpus in the square $[i, j]$.
- Let $S_{i,j}$ be true if agent perceives stench in the square $[i, j]$.
- Let $V_{i,j}$ be true if that square $[i, j]$ is visited.
- Let $G_{i,j}$ be true if there is gold (and glitter) in the square $[i, j]$.
- Let $OK_{i,j}$ be true if the room is safe.



Atomic Proposition Variable for Wumpus World



1,4	2,4 P?	3,4	4,4
1,3 W?	2,3 S G B	3,3	4,3
1,2	2,2 V P?	3,2	4,2
1,1 A ok	2,1 B V ok	3,1 P?	4,1

$$(R1) \neg S_{11} \rightarrow \neg W_{11} \wedge \neg W_{12} \wedge \neg W_{21}$$

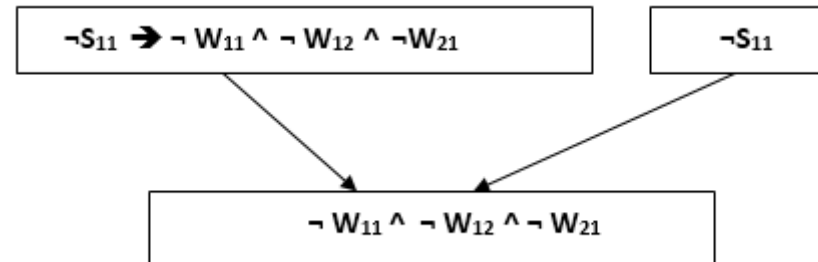
$$(R2) \neg S_{21} \rightarrow \neg W_{11} \wedge \neg W_{21} \wedge \neg W_{22} \wedge \neg W_{31}$$

$$(R3) \neg S_{12} \rightarrow \neg W_{11} \wedge \neg W_{12} \wedge \neg W_{22} \wedge \neg W_{13}$$

$$(R4) S_{12} \rightarrow W_{13} \vee W_{12} \vee W_{22} \vee W_{11}$$

Prove that Wumpus is in the room (1, 3)

- We will firstly apply MP rule with R1 which is $\neg S_{11} \rightarrow \neg W_{11} \wedge \neg W_{12} \wedge \neg W_{21}$, and $\neg S_{11}$ which will give the output $\neg W_{11} \wedge \neg W_{12} \wedge \neg W_{21}$.



Apply And-Elimination Rule:

- After applying And-elimination rule to $\neg W_{11} \wedge \neg W_{12} \wedge \neg W_{21}$, we will get three statements:
 $\neg W_{11}$, $\neg W_{12}$, and $\neg W_{21}$.
- Modus Ponens will be applied on other states and the unit Resolution to reach goal

Logical Agents

- Logical agents are a type of intelligent agent in artificial intelligence that **use logical reasoning to make decisions and take actions**. These agents **operate based on a knowledge base** represented in some form of **logical language**, such as **propositional logic** or **first-order logic**. They typically follow a cycle of perceiving their environment, updating their knowledge base with new information, reasoning about that information, and then selecting actions based on the conclusions of their reasoning.
- Key components of logical agents include:
 1. **Knowledge Base:** This is the **set of logical sentences** that represent the agent's knowledge about the world. It includes both explicit knowledge (**facts** that the agent knows) and implicit knowledge (**inferences** that the agent can make).
 2. **Inference Mechanism:** Logical agents **use rules of inference to derive new information** from their knowledge base. These rules may include deductive reasoning (e.g., modus ponens, resolution) and possibly other forms of inference such as abduction or induction.
 3. **Decision Making:** Once the agent has inferred new information from its knowledge base, it must decide on an **action** to take **based on its goals and the current state of the world**. This decision-making process often involves selecting actions that are consistent with the agent's beliefs and preferences.
 4. **Execution:** After selecting an action, the agent **executes it in the environment and observes the resulting state**, which may lead to further updates to its knowledge base.
- Logical agents are particularly useful in domains where reasoning about complex relationships and making decisions based on logical rules is important, such as expert systems, theorem proving, natural language understanding, and planning. However, they are limited by the expressiveness and scalability of the logical formalisms they use and may struggle with uncertainty and incomplete information.

Knowledge Base

A knowledge base in the context of logical agents is formed by gathering and **representing information** about the world in a **structured form** that can be reasoned about logically. Here's how a knowledge base is typically formed:

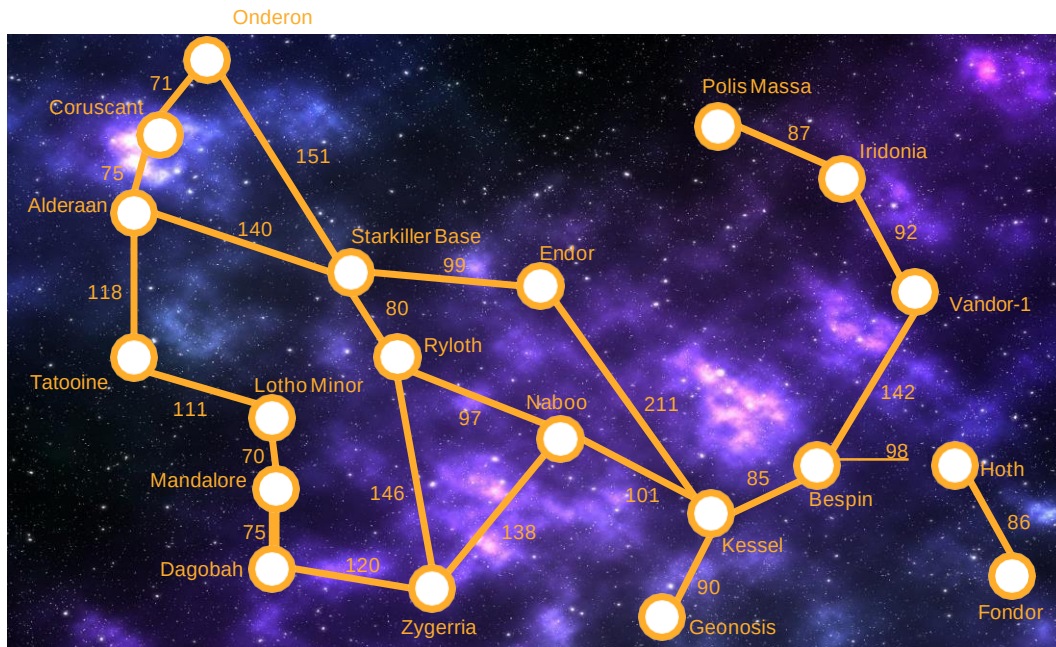
1. **Knowledge Acquisition:** The process of acquiring knowledge involves gathering information about the domain in which the agent operates. This information can come from various sources, such as domain experts, textbooks, databases, or through observation and experience. **Knowledge acquisition can be manual**, where human experts input information into the system, or automated, where the system gathers data from sources like databases or the internet.
2. **Representation:** Once the knowledge is acquired, it needs to be represented in a form that the agent can understand and reason about. In the context of logical agents, this representation is typically in the form of **logical statements or propositions**. The knowledge base may include facts about the domain, rules or constraints governing the domain, and background knowledge necessary for reasoning.
3. **Formalization:** The acquired knowledge is formalized into a logical language, such as propositional logic, first-order logic, or higher-order logic. In propositional logic, the knowledge base consists of **atomic propositions that are either true or false**. In **first-order logic**, the knowledge base includes **predicates, variables, quantifiers, and logical connectives** to represent relationships and rules in the domain.
4. **Maintenance and Updates:** The knowledge base is not static; it needs to be **updated and maintained** as new information becomes available or the state of the world changes. This includes adding new facts, modifying existing rules, or retracting information that is no longer valid. The **agent may also derive new knowledge through inference** and add it to the knowledge base.
5. **Validation and Verification:** Before deploying the knowledge base in an agent, it's essential to validate and verify its correctness and consistency. This involves **testing the knowledge base against known scenarios** or using formal methods to ensure that it accurately represents the domain and follows the rules of logical inference.

By following these steps, a logical agent can form a knowledge base that serves as the foundation for its reasoning and decision-making processes in its environment.

Knowledge-based Agents

Knowledge-based agents use a process of **reasoning** over an internal **representation** of knowledge to decide what action to take.

So far, our **problem-solving** agents have performed a **search** over **states** in order to find a plan. The representation of states has been **atomic**.



Limited to commands like
“Navigate to Kessel”

“Take me to the nearest
habitat planet where I can store
my perishable cargo”

Knowledge-based Agents

A central component of a knowledge-based agent is a **knowledge base** or KB.

A KB contains a set of **sentence** that are written in a **knowledge representation language**. The sentence contains some assertion about the world.

Natural language sentences	Knowledge representation language sentence
<i>Hoth is a planet</i>	<code>planet(hoth)</code>
<i>Hoth is habitable</i>	<code>habitable(hoth)</code>
<i>Hoth is far from itssun</i>	<code>far_from(hoth, sol)</code>
<i>If a planet is far from its sun then it is cold</i>	<code>planet(x) and sun(y) and far_from(x,y) → cold(x)</code>

Knowledge-based Agents

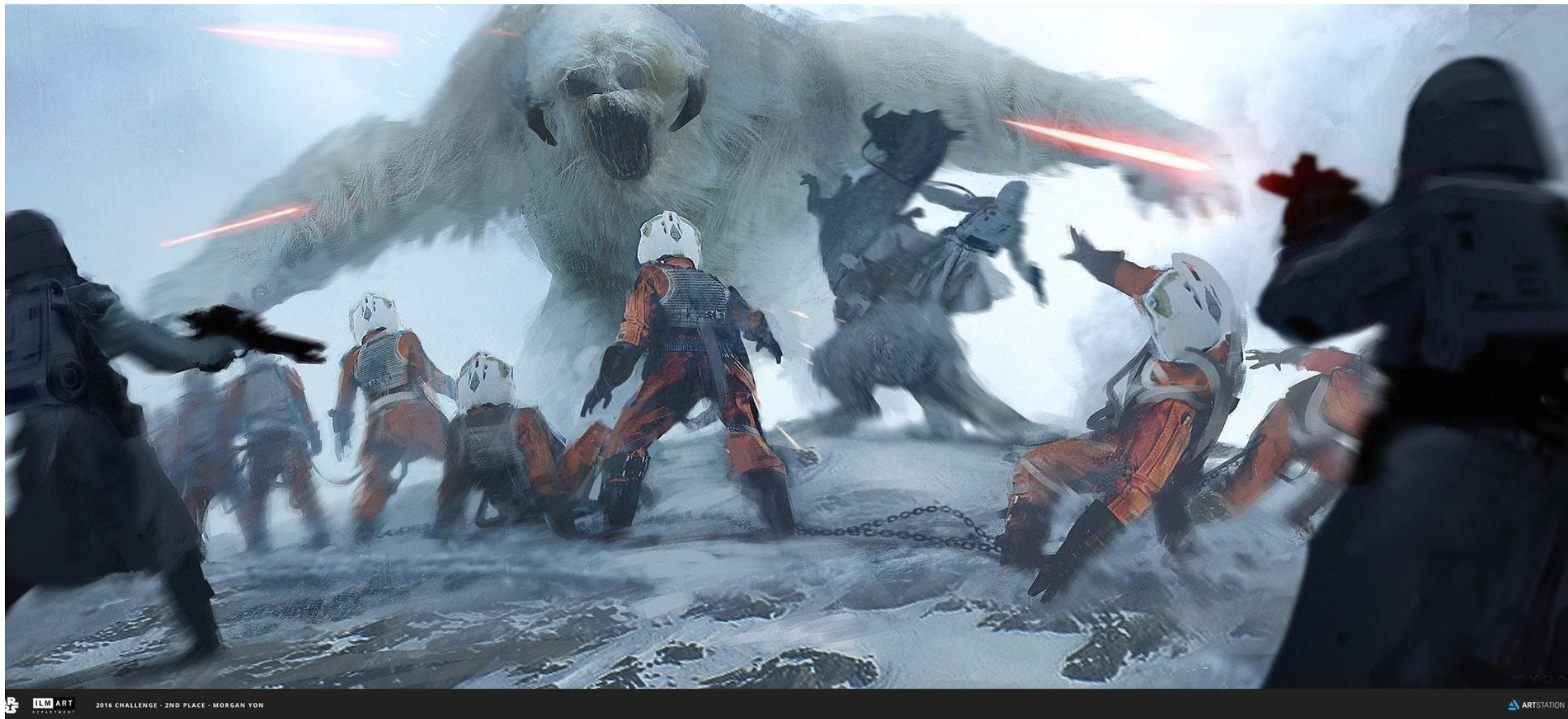
There are two kinds of sentences:

- **Axioms** – a sentence that is given
- **Derived sentences** – a new sentence that is derived from others sentences The process of deriving new sentences from old sentences is called **inference**.

A KB can initially contain some **background information** about the world, and a knowledge-based agent can add to the information in the KB through its observations of the world.

In addition to asserting new knowledge into its KB, a knowledge-based agent can also query the KB and ask it to derive new knowledge in order to select what action it should take.

Hunt the Wampas



Wampa World

Our **knowledge-based agent**, R2D2, explores a **cave** consisting of **rooms** connected by passageways.

Lurking somewhere in the cave is the **Wampa**, a beast that eats any agent that enters its room.

Some rooms contain bottomless **pits** that trap any agent that wanders into the room.

In one room is master **Luke**.

The goal is:

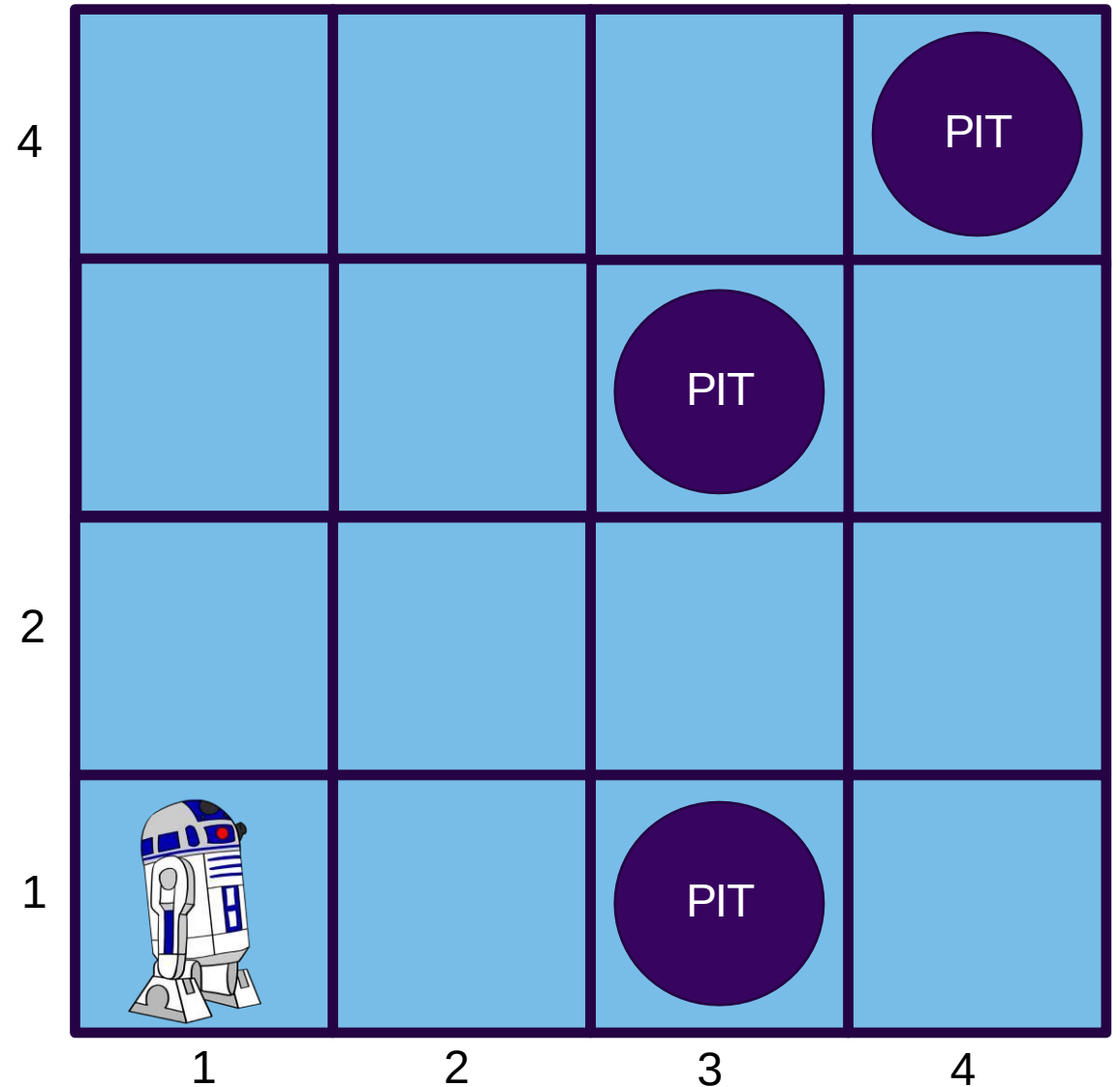
- collect Luke
- exit the world
- without being eaten



Wampa World

Environment:

- A 4x4 grid of rooms. The agent starts in the square [1,1].
- Wampa and Luke are randomly placed in other squares.
- Each square can be pit with 20% probability.



Wampa World

Performance measure:

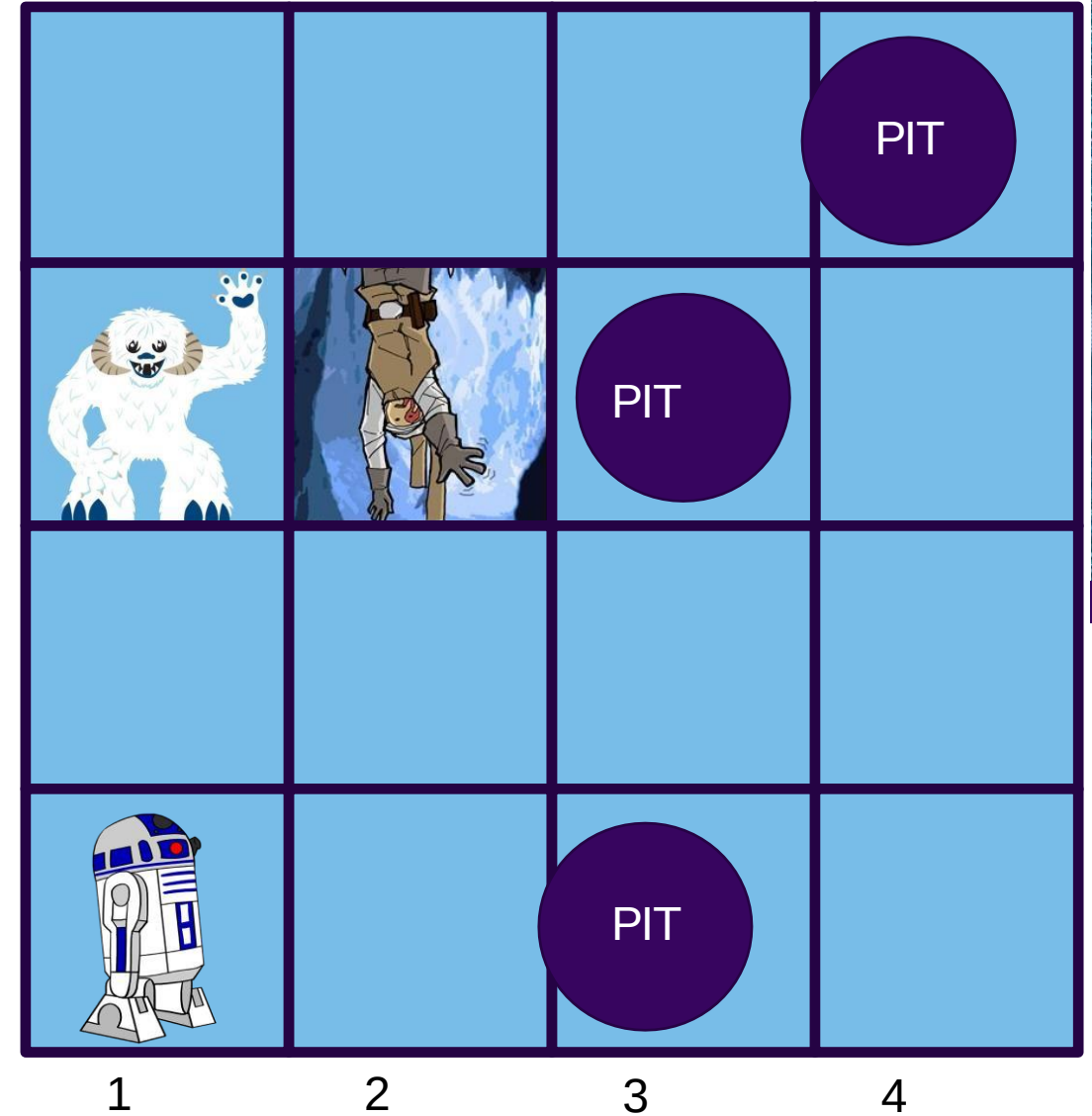
- +1000 points for rescuing Luke and leaving the cave
- -1000 for falling into a pit or being eaten by the Wampa
- -1 for each action taken
- -10 for using up your blaster fire

4

3

2

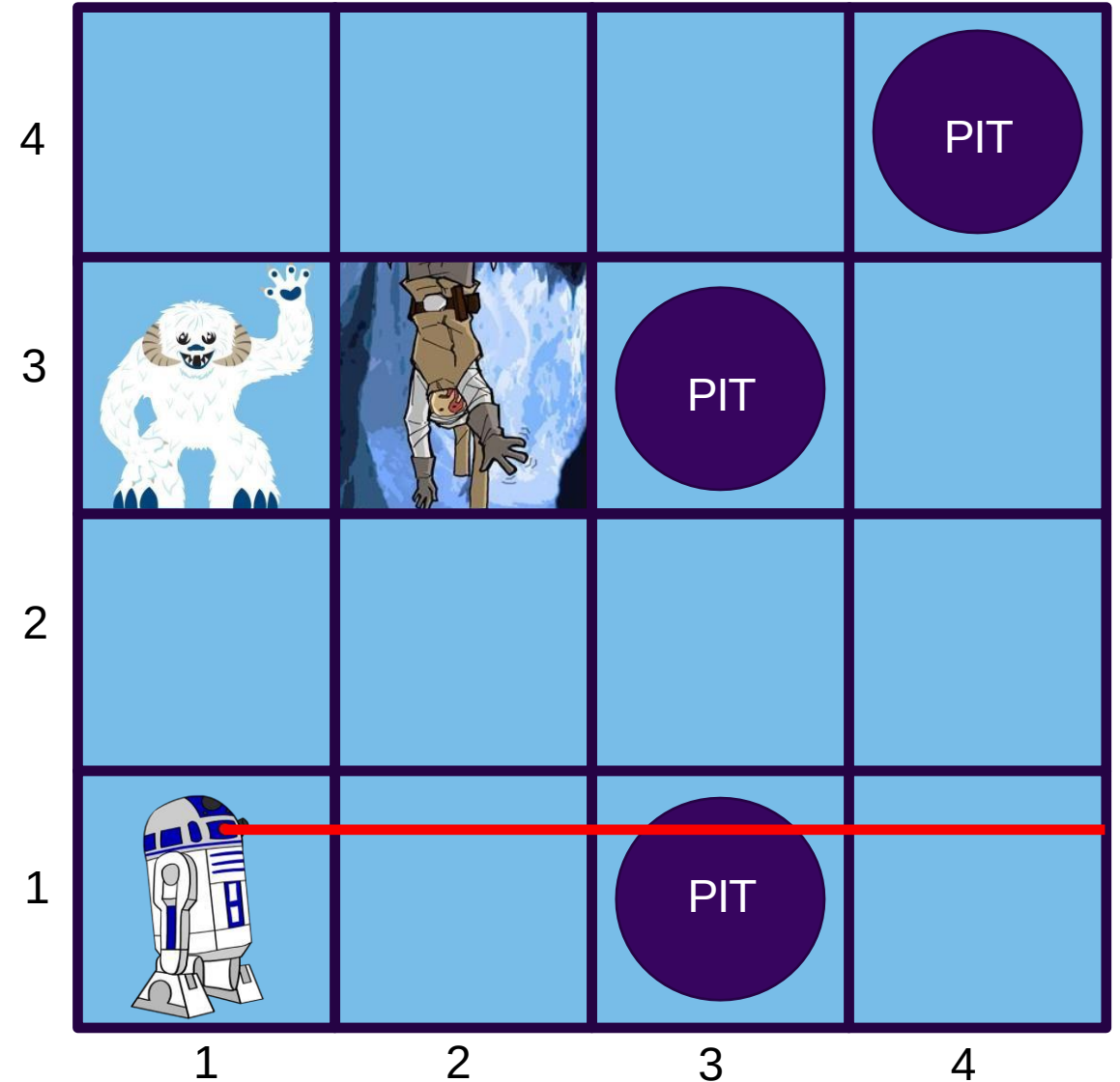
1



Wampa World

Actuators:

- R2 can move *Forward*, *TurnLeft*, *Turn right*.
- Agent dies if it moves into a pit or a Wumpus square.
- *Grab* can pick up Luke.
- *Shoot* fires blaster bolt in a straight line in the direction that R2D2 is facing.
- If the blaster hits the Wampa, it dies. R2 only has enough power for one shot.
- *Climb* gets R2 out of the cave but only works in [1, 1]

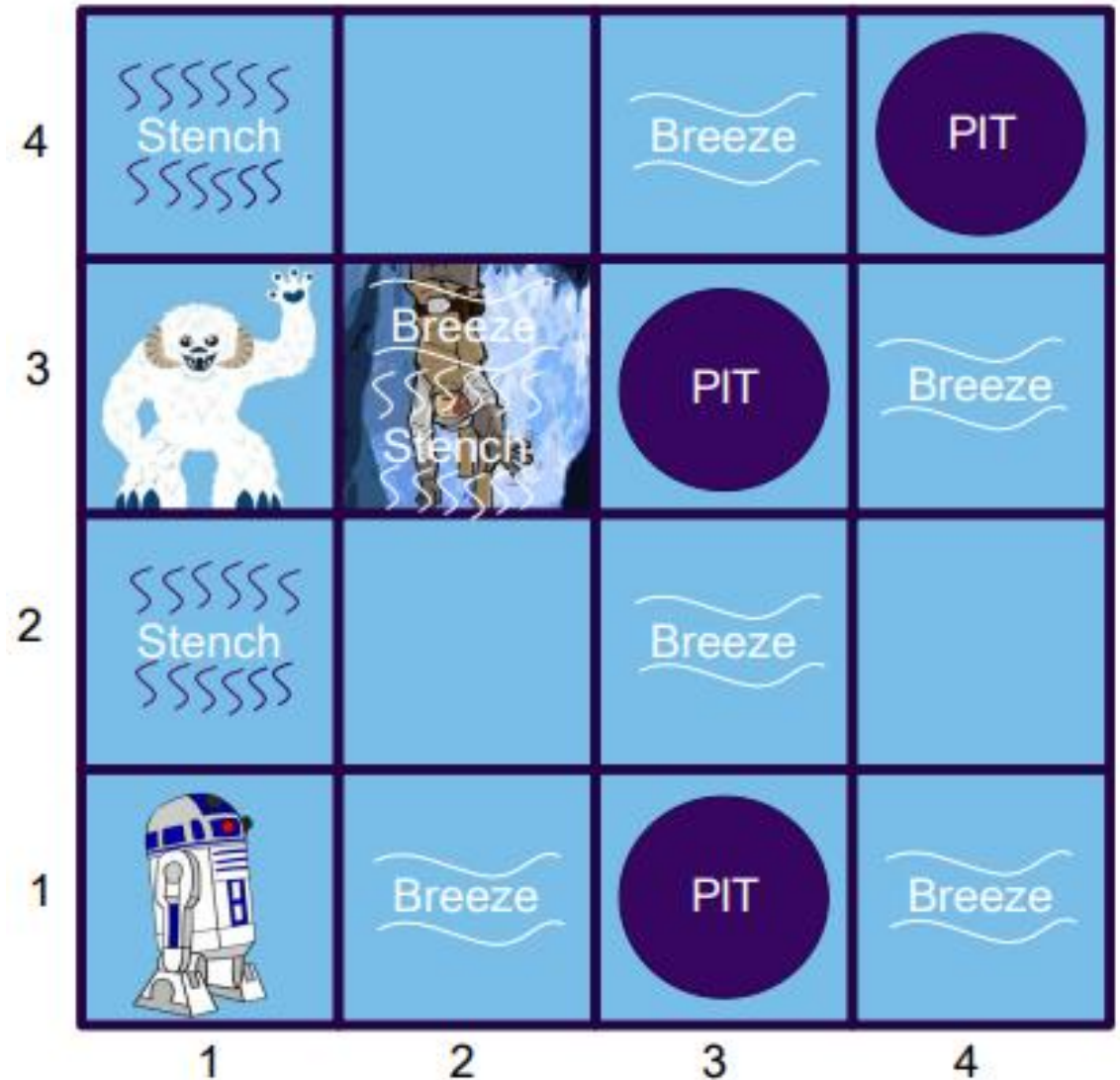


Wampa World

Sensors:

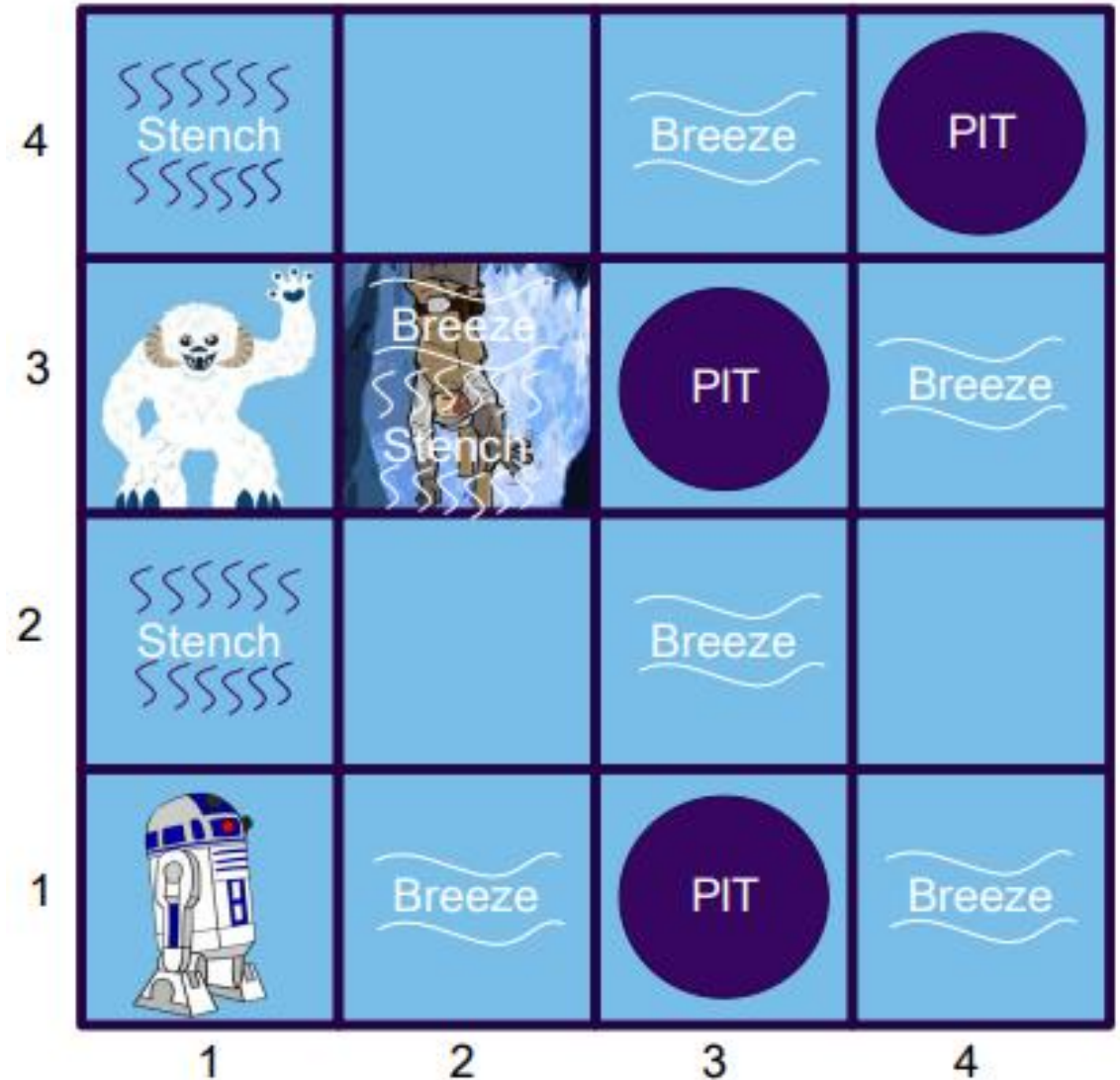
- In each square adjacent to the Wampa, R2D2's olfactory sensor perceives a *Stench*
- In each square adjacent to a pit, R2D2's wind sensor perceives a *Breeze*
- In the square with Luke, R2D2's audio sensor perceives a *Gasp*
- When R2D2 walks into a wall it perceives a *Bump*
- When the Wampa is killed, R2D2's audio sensor perceives a *Scream*

Percept = [*Stench*, *Breeze*, *None*, *None*, *None*]



Wampa World

1. **Deterministic** (outcomes are specified)
2. **Discrete**
3. **Single-agent**
4. **Static** (Wampa and pits doesn't move)
5. **Sequential** because reward doesn't come for many steps
6. **Partially observable** because some parts of the state are not directly perceivable: Location of Luke, Wampa, and pits aren't directly observable, only local perception.

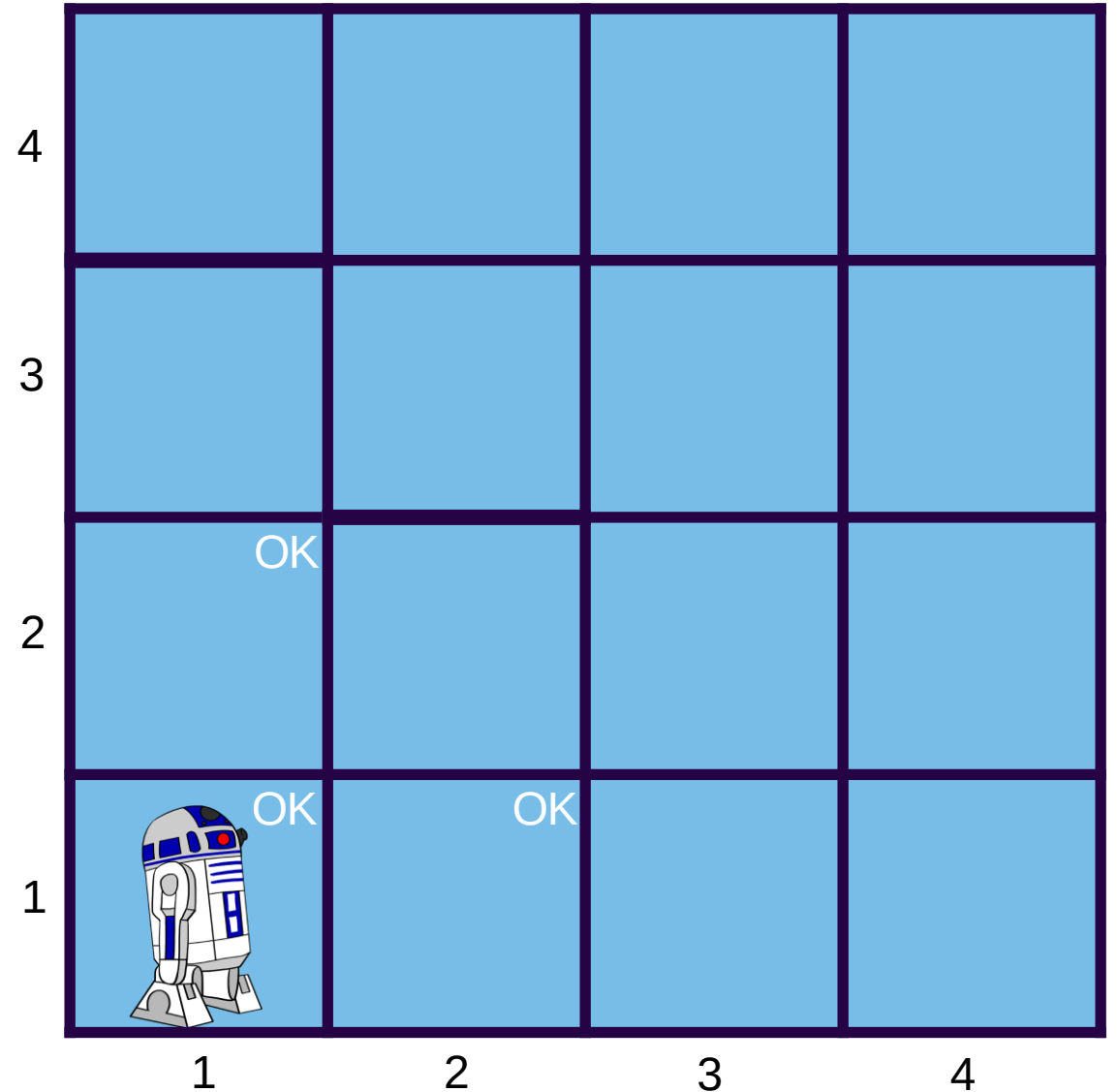


Wampa World Walkthrough

R2D2 starts in [1,1]

Percept=[*None, None, None, None, None*]

What can we conclude about [1,2] and [2,1]?

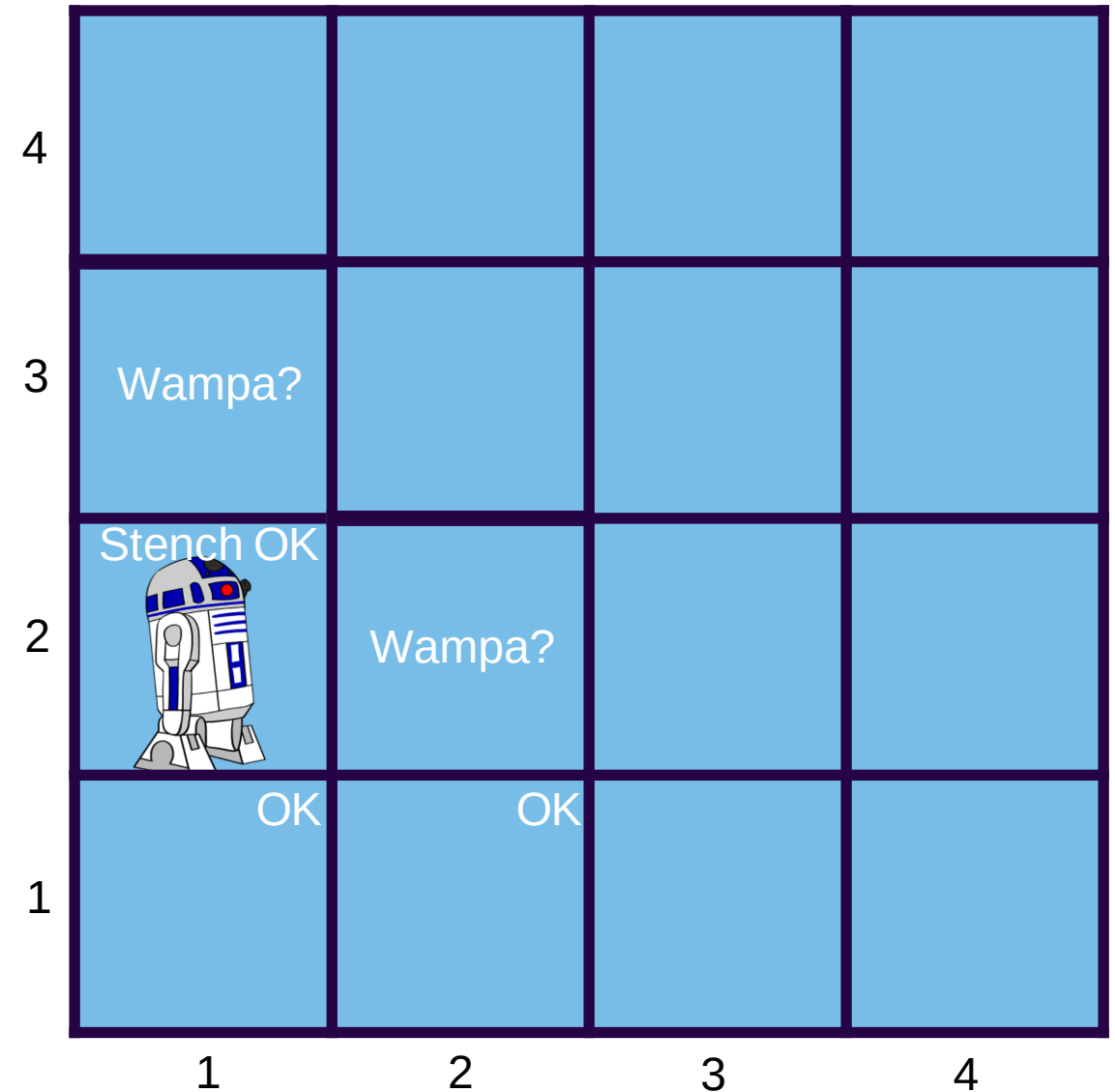


Wampa World Walkthrough

R2D2 moves to [1,2]

Percept=[*Stench*, *None*, *None*, *None* *None*]

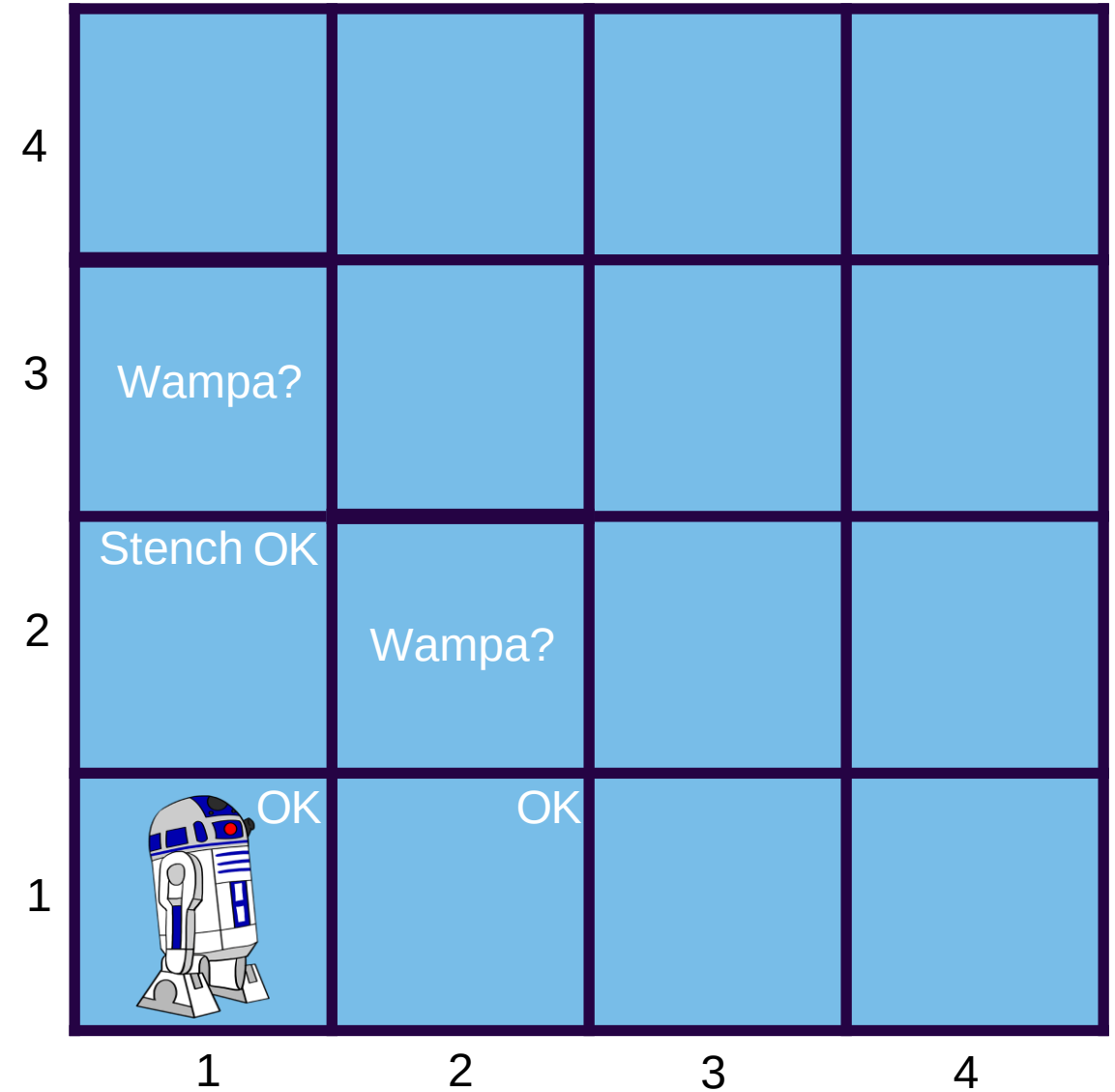
What can we conclude about [3,1] and [2,2] from the Stench?



Wampa World Walkthrough

R2D2 moves back to [1,1] and gets the same percept vector as before

Percept=[*None, None, None, None None*]

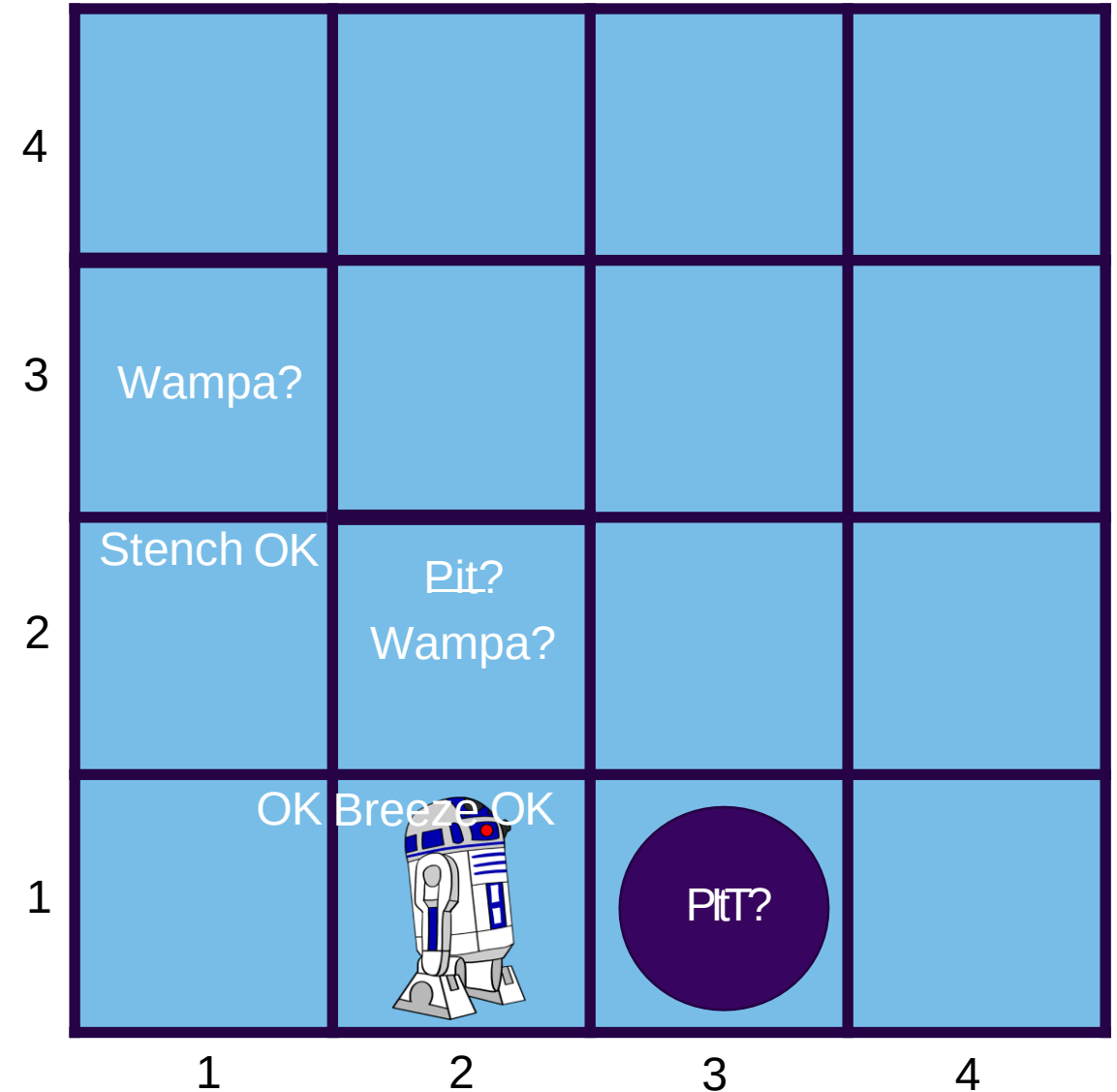


Wampa World Walkthrough

R2D2 moves to [2,1]

Percept=[*None, Breeze, None, None None*]

What can we conclude about [3,1] and [2,2] based on the Breeze?

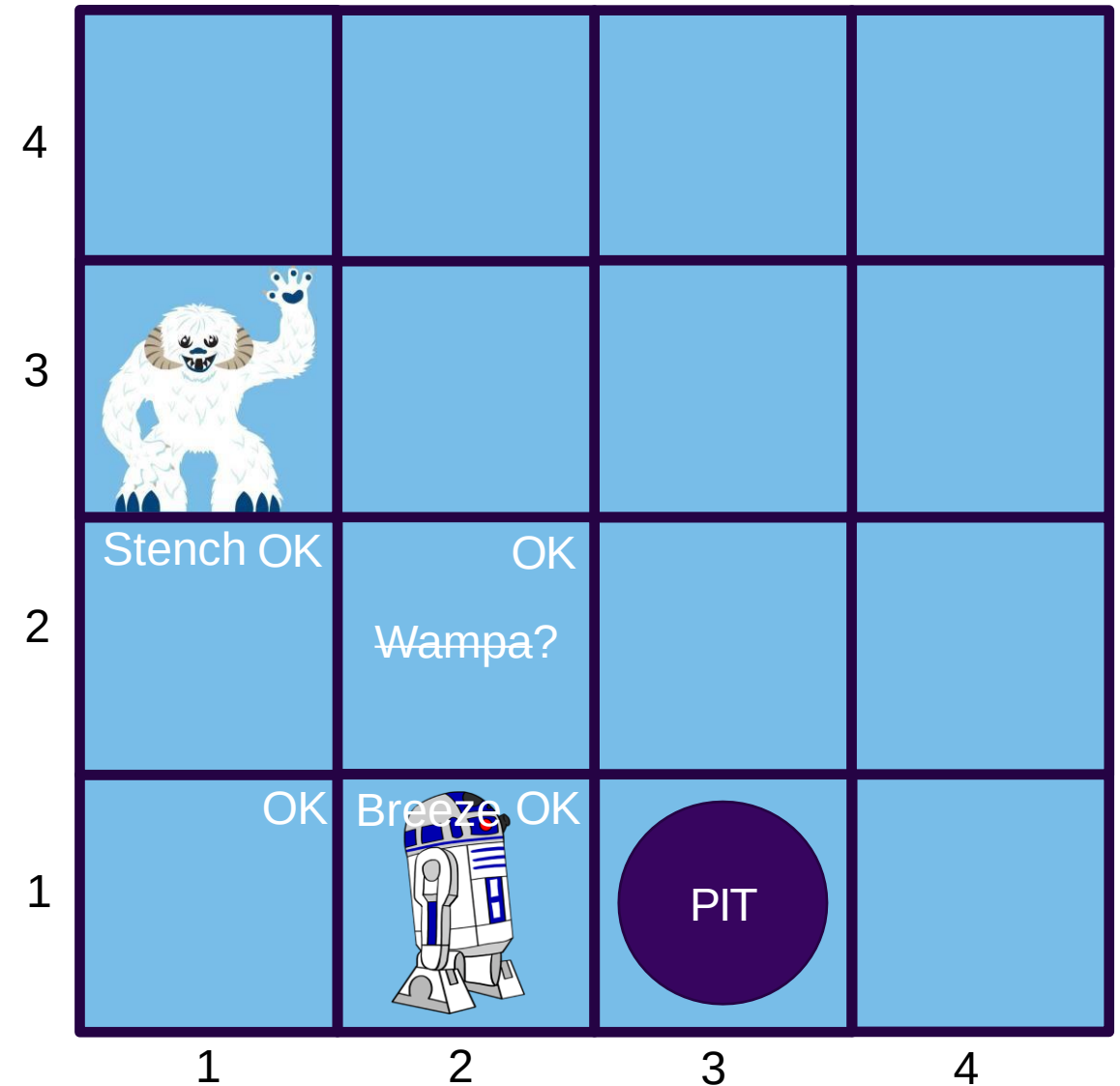


Wampa World Walkthrough

R2D2 moves to [2,1]

Percept=[*None, Breeze, None, None None*]

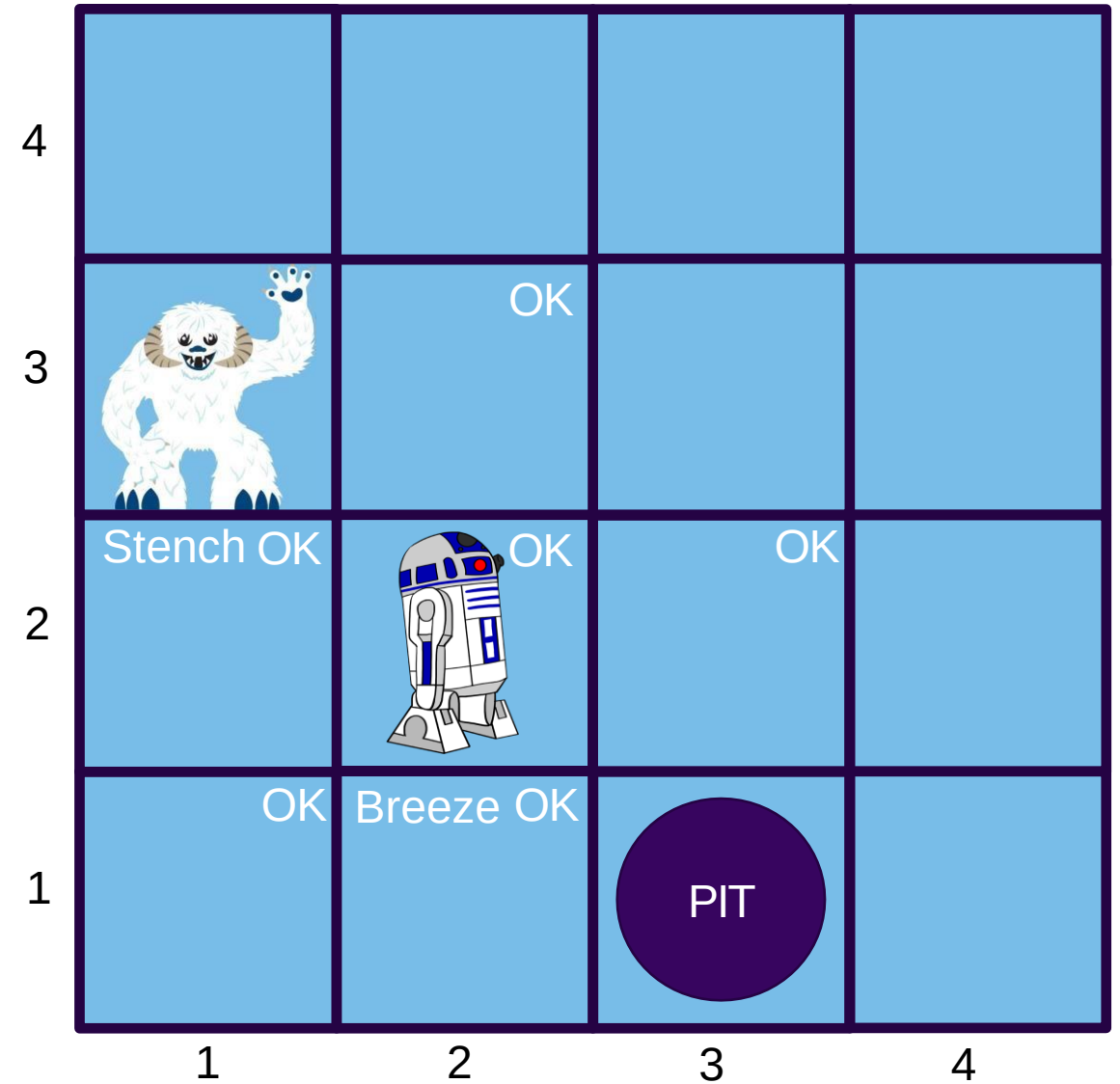
What can we conclude about [2,2] and [1,3] based on the lack of a Stench here?



Wampa World Walkthrough

R2D2 moves to [2,2]

Percept=[*None, None, None, None, None*]

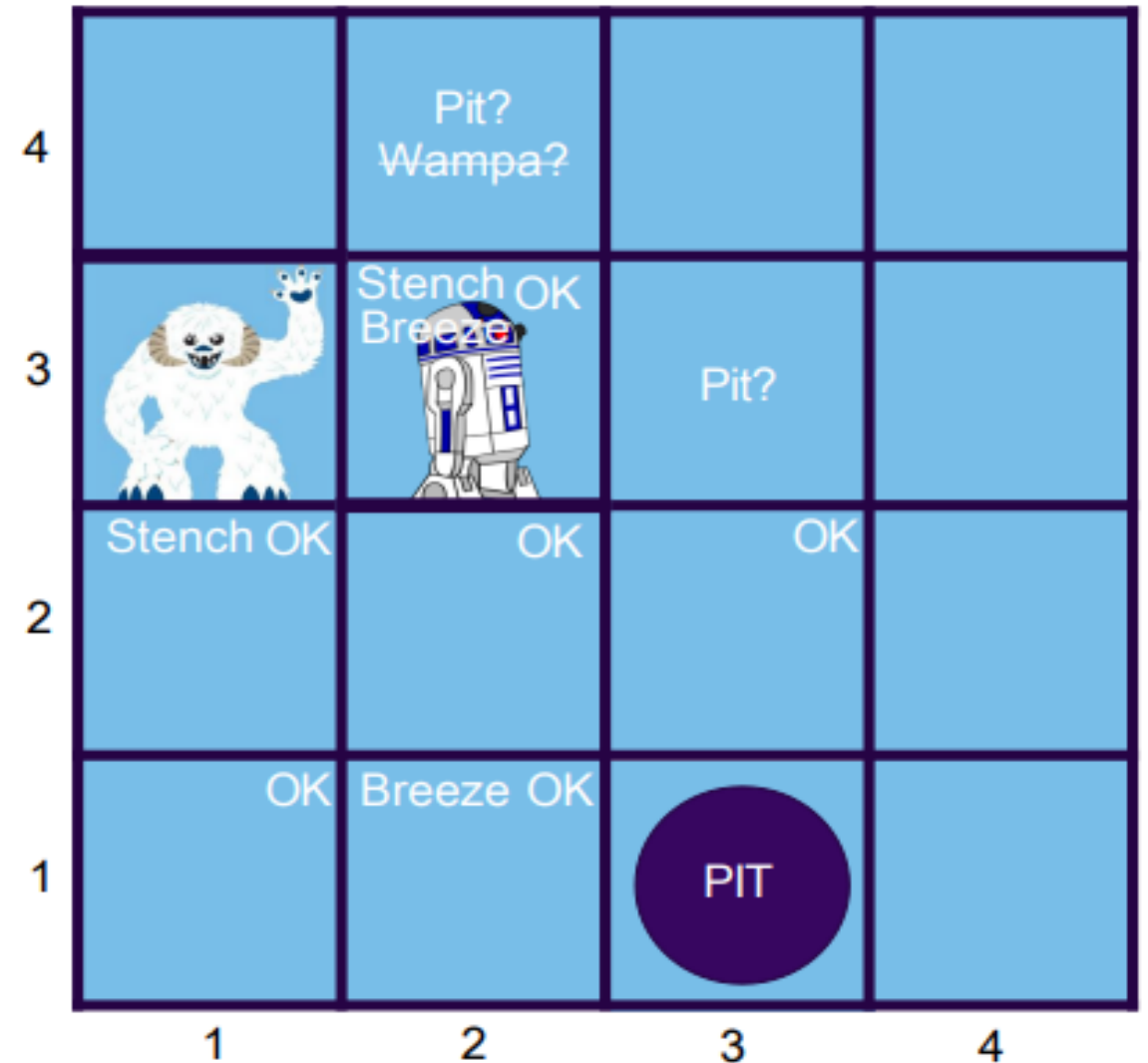


Wampa World Walkthrough

R2D2 moves to [2,3]

Percept=[*Stench*, *Breeze*, *Gasp*, *None*, *None*]

What can we conclude about [2,4] or [3,3]?

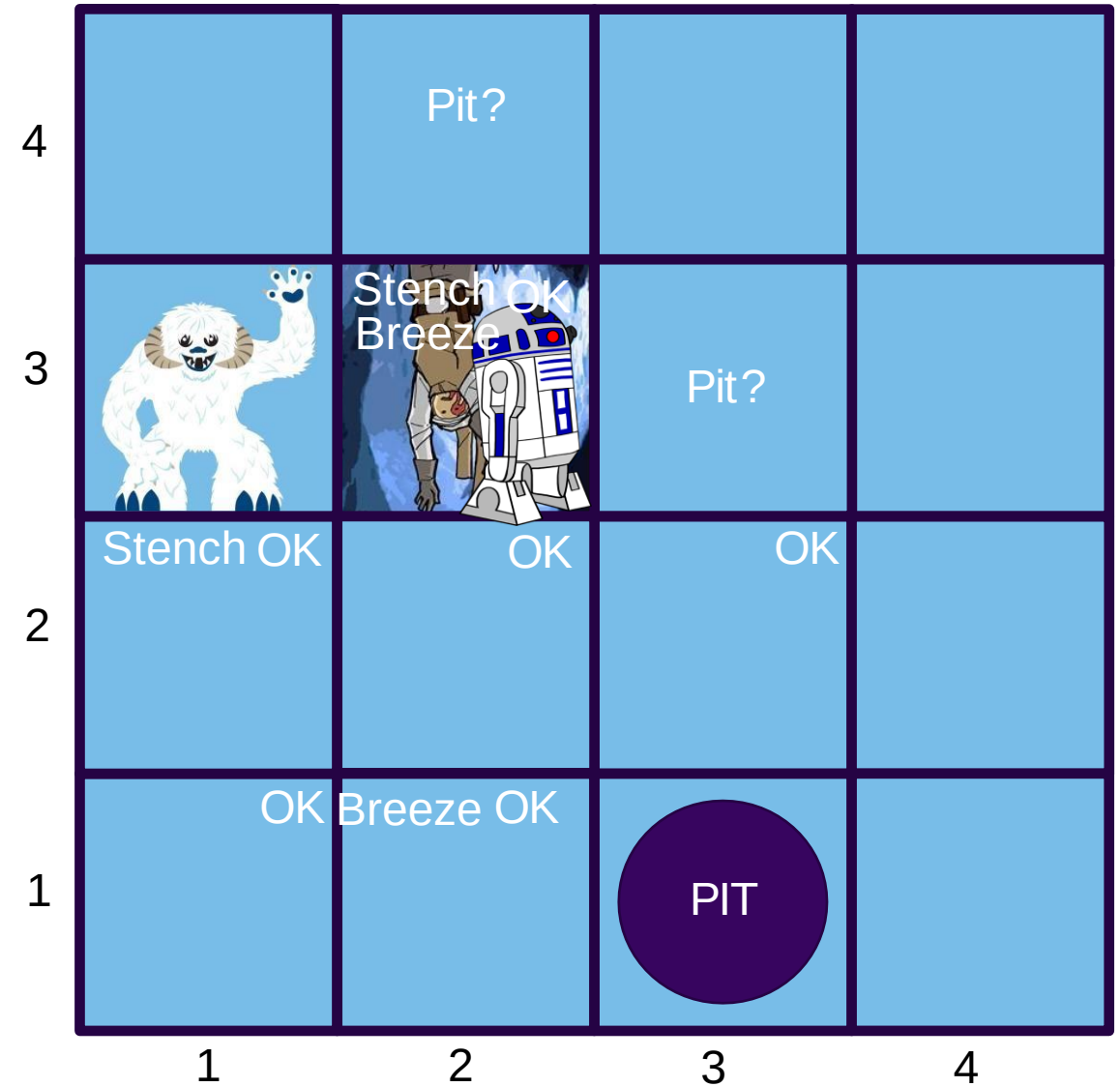


Wampa World Walkthrough

R2D2 moves to [2,3]

Percept=[*Stench*, *Breeze*, *Gasp*, *None*, *None*]

We heard a Gasp, so Luke is here!



Logic

Logic can serve as a general class of **representations** for knowledge-based agents. Here we are going to examine **Propositional Logic**.

- KB consists of **sentences** in the **representation language**
- Representation language has a **syntax** that specifies sentences that are well-formed
- A logic defines the **semantics** of the sentences, which is their **meaning**
- The semantics defines the **truth** of each sentence with respect to a **possible world**, which we will often call a **model**.

Possible worlds and models

- Models are mathematical abstractions that have a fixed set of **truth values** which are **{true, false}** for each sentence.
- If sentence α is true in model m then we say
 - m satisfies α , or
 - m is a model of α
- We use the notation $M(\alpha)$ to mean the **set of all models** of α .

For instance, α could be a sentence that means “there is no pit in [2,2]”. In that case, $M(\alpha)$ would be all instances of Wampa World where [2,2] doesn’t have a pit.

Logical Entailment

Once we have a notion of truth, we can start to define **logical reasoning**. Logical reasoning involves the **entailment** relation between sentence.

In plain English, entailment is the idea that a sentence **follows logically** from another sentence.

To write sentence α entails sentence β in mathematical notation we use the $\models$ symbol:

$$\alpha \models \beta$$

The definition is

$$\alpha \models \beta \text{ if and only if } M(\alpha) \subseteq M(\beta)$$

This means that α is more specific, or stronger than, β . For instances, β could mean that “The agent is a robot” and α could mean “The agent is an astromech”.

Knowledge Base

The KB can be thought of as a set of sentences.

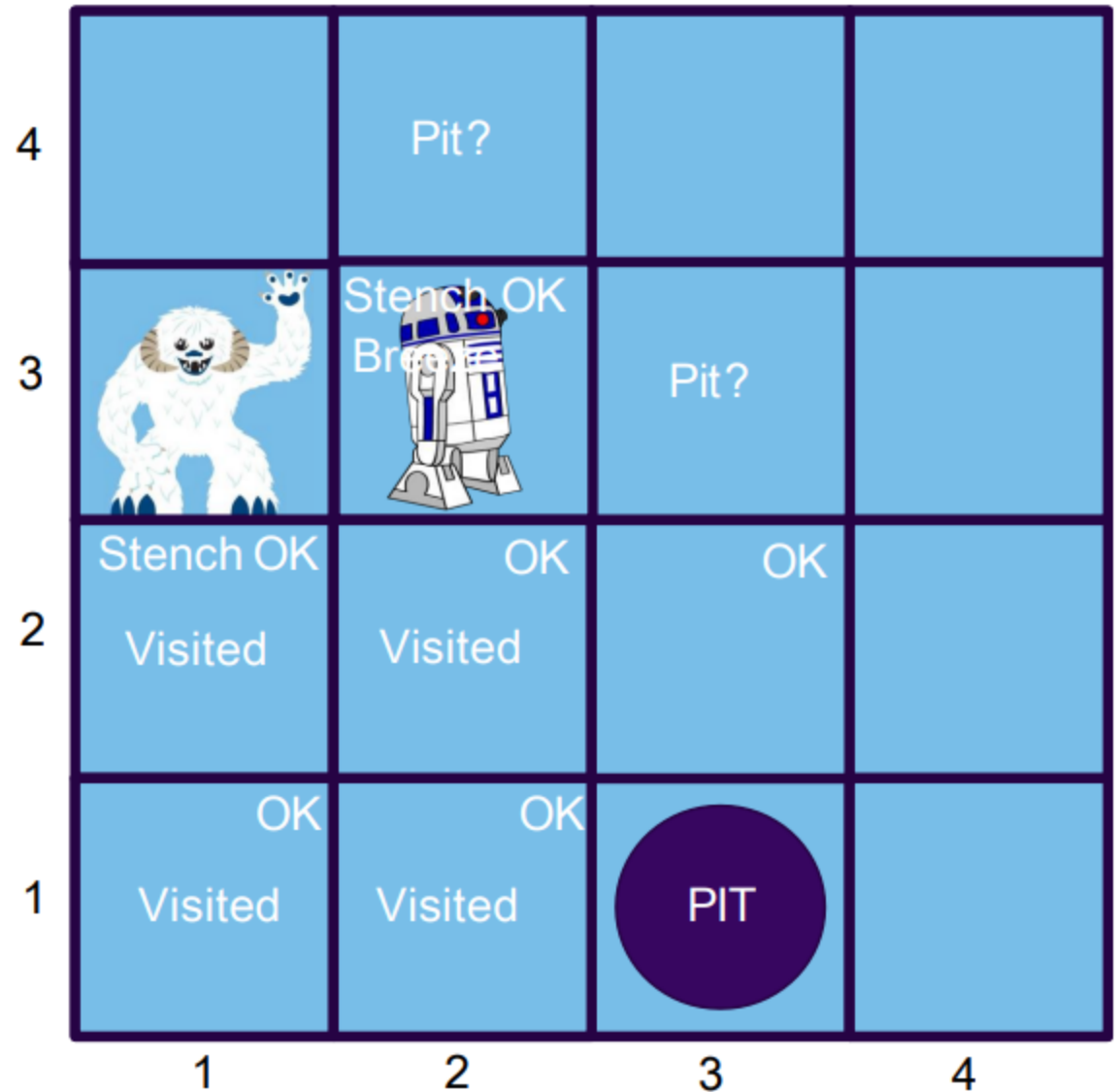
α_1 = "There is no pit in [1,2]"

α_2 = "There is a pit in [3,1]"

α_3 = "There is a wampa in [1,3]"

The KB is false in models that contradict what the agent knows. For example, the KB is false in any model m where [1,2] contains a pit.

Possible Worlds is the process of enumerating all Possible Worlds that are compatible with the KB. $M(KB) \subseteq M(\alpha_1)$



Logical inference

Entailment can be applied to derive conclusions, which is the process of **logical inference**.

We can think about the consequences of a KB as a large set of additional sentences that are entailed given the sentences that have been added to the KB.

We would like to design **inference algorithms** to enumerate these sentences.

When an inference algorithm **i** allows us to conclude that α is true, then we write

$$\text{KB} \vdash_i \alpha$$

“ α is derived from **KB** by **i**”

Propositional Logic

Propositional Logic

Atomic sentences are represented with a single **propositional symbols**.
Propositional symbols **stand for a statement** that can be true or false.

For example, $W_{1,3}$ is a propositional symbol that we choose to stand for
“There is a Wampa at location[1,3]”

That statement can be true or false.

The symbol **FacingEast** could stand for “The agent is currently facing East”.

Propositional Logic

Complex sentences are constructed from simpler ones using parentheses and **logical connectives**.

Logical Connective	Meaning
$\neg$ (not)	$\neg W_{1,3}$ is the negation of $W_{1,3}$
$\wedge$ (and)	$W_{1,3} \wedge P_{3,1}$ is called a conjunction
$\vee$ (or)	$W_{1,3} \vee P_{3,1}$ is called a disjunction
$\Rightarrow$ (implies)	$W_{1,3} \Rightarrow S_{1,2}$ is called an implication. $W_{1,3}$ is its premise or antecedent and $S_{1,2}$ is its conclusion or consequence
- (if and only if)	$W_{1,3} \Leftrightarrow \neg W_{3,4}$ is called an biconditional

Truth Tables

Negation	
P	$\neg P$
True	False
False	True

"It is not the case that the Death Star is a moon" is **true** because "the Death Star is a moon" is **false**.

"It is not the case that Wampas smell bad" is **false** because "Wampas smell bad" is **true**.

Truth Tables

Conjunction		
P	Q	$P \wedge Q$
True	True	True
True	False	False
False	True	False
False	False	False

P and Q are true:

“Wampas smell bad **and** Tauntauns smell bad.”

(This sentence is true)

P is true and Q is false:

“Wampas smell bad **and** Tauntauns are robots.”

(This sentence is false)

P is false and Q is false:

“Wampas smell good **and** Tauntauns are robots.”

(This sentence is false)

Truth Tables

Disjunction		
P	Q	$P \vee Q$
True	True	True
True	False	True
False	True	True
False	False	False

P and Q are true:

“Wampas smell bad **or** Tauntauns smell bad.” (This sentence is true)

P is true and Q is false:

“Wampas smell bad **or** Tauntauns are robots.” (This sentence is true)

P is false and Q is false:

“Wampas smell good **or** Tauntauns are robots.” (This sentence is false)

Truth Tables

Conditional		
P	Q	$P \Rightarrow Q$
True	True	True
True	False	False
False	True	True
False	False	True

Which of these scenarios did break my promise to you?

This one.

To understand why the conditional is defined this way assume that I tell you this $P \Rightarrow Q$:
If you join the dark side then we will rule the galaxy together.

In which of these four scenario did I tell a lie?

1. **You join the dark side, and we rule the galaxy together.** (Both P and Q are True)
2. You join the dark side, **but** we **don't** rule the galaxy together. (P is True, Q is False)
3. You **don't** join the dark side, **but** we **still** rule the galaxy together. (P is False, Q is True)
4. You **don't** join the dark side, **and** we **don't** rule the galaxy together. (P is False, Q is False)

Truth Tables

Shorthand for
 $P \Rightarrow Q \wedge Q \Rightarrow P$

Biconditional		
P	Q	$P \Leftrightarrow Q$
True	True	True
True	False	False
False	True	False
False	False	True

I tell you: The Death Star can be destroyed **if and only if** your missile hits its vulnerable spot.

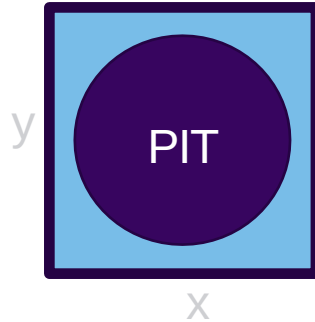
1. **The Death Star is destroyed, and you hit the vulnerable spot.** (Both P and Q are True)
2. The Death Star **is** destroyed, **but** you **didn't** hit its vulnerable spot. (P is True, Q is False)
3. The Death Star **isn't** destroyed, **but** you **did** hit its vulnerable spot. (P is False, Q is True)
4. The Death Star **isn't** destroyed, **and** you **also didn't** hit its vulnerable spot. (P is False, Q is False)

Knowledge Bases

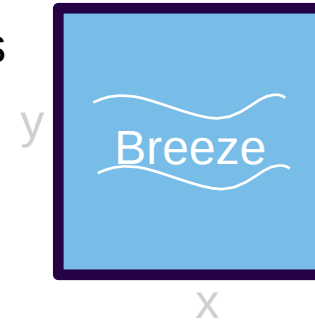
A Simple Knowledge Base

We can construct a knowledge base for Wampa World.
Let's start with a set of symbols for each location $[x,y]$:

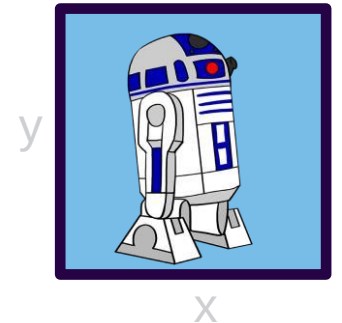
$P_{x,y}$ is true if there is
a pit in $[x,y]$



$B_{x,y}$ is true if there is
a breeze in $[x,y]$



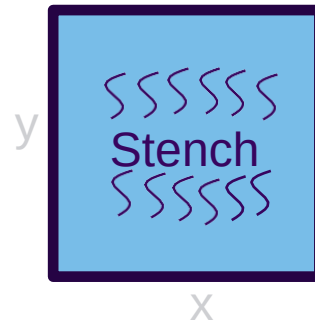
$L_{x,y}$ is true
if the agent
is in $[x,y]$



$W_{x,y}$ is true if there is
a Wampa in $[x,y]$



$S_{x,y}$ is true if there is
a stench in $[x,y]$



A Simple Knowledge Base

We can construct sentences out of these using our logical connectors. We'll label each sentence.

R1: $\neg P_{1,1}$

R2: $B_{1,1} \Leftrightarrow (P_{1,2} \vee P_{2,1})$

R3: $B_{2,1} \Leftrightarrow (P_{1,1} \vee P_{2,2} \vee P_{3,1})$

These are true of all Wampa Worlds.

Logical Connectives

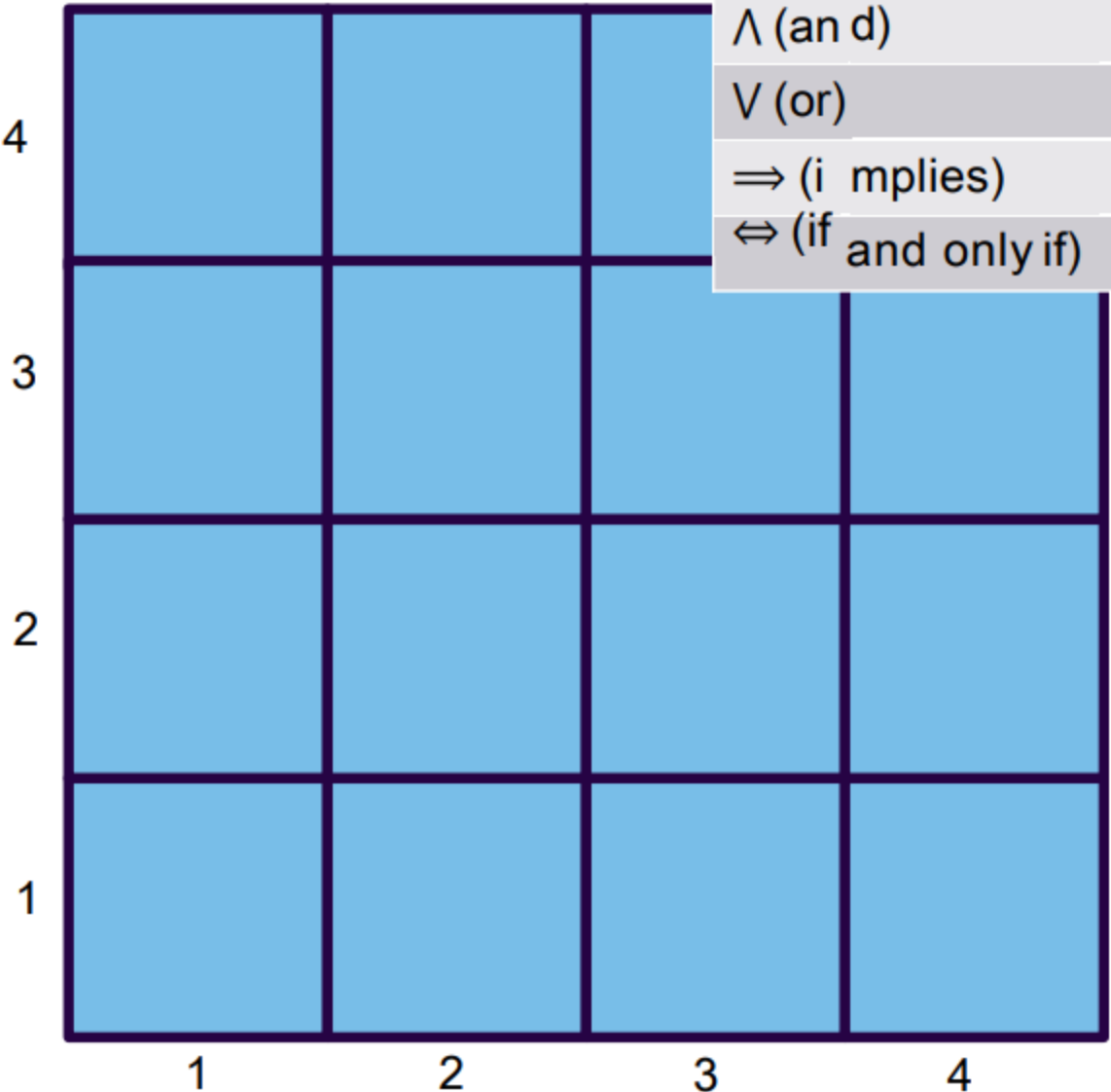
$\neg$ (not)

$\wedge$ (and)

$\vee$ (or)

$\Rightarrow$ (implies)

$\Leftrightarrow$ (if and only if)



A Simple Knowledge Base

We can construct sentences out of these using our logical connectors. We'll label each sentence.

R1: $\neg P_{1,1}$

R2: $B_{1,1} \Leftrightarrow (P_{1,2} \vee P_{2,1})$

R3: $B_{2,1} \Leftrightarrow (P_{1,1} \vee P_{2,2} \vee P_{3,1})$

What if we perceive the presence or absence of breeze in $[1,1], [2,1]$?

R4: $\neg B_{1,1}$

R5: $B_{2,1}$

Logical Connectives

$\neg$ (not)

$\wedge$ (and)

$\vee$ (or)

$\Rightarrow$ (implies)

$\Leftrightarrow$ (if and only if)

3				
2				
1	No Breeze	Breeze		
	1	2	3	4

A Simple Knowledge Base

We can construct sentences out of these using our logical connectors. We'll label each sentence.

R1: $\neg P_{1,1}$

R2: $B_{1,1} \Leftrightarrow (P_{1,2} \vee P_{2,1})$

R3: $B_{2,1} \Leftrightarrow (P_{1,1} \vee P_{2,2} \vee P_{3,1})$

R4: $\neg B_{1,1}$

R5: $B_{2,1}$

Can we mechanically combine the sentences in our KB to prove that a pit exists at (or is absent from) any location?

Logical Connectives

$\neg$ (not)

$\wedge$ (and)

$\vee$ (or)

$\Rightarrow$ (implies)

$\Leftrightarrow$ (if and only if)

KB $\models \neg P_{1,2}$
Does the KB entail that there is no pit here?

2

Pit?

Pit?

1

No Breeze

Breeze

Pit?

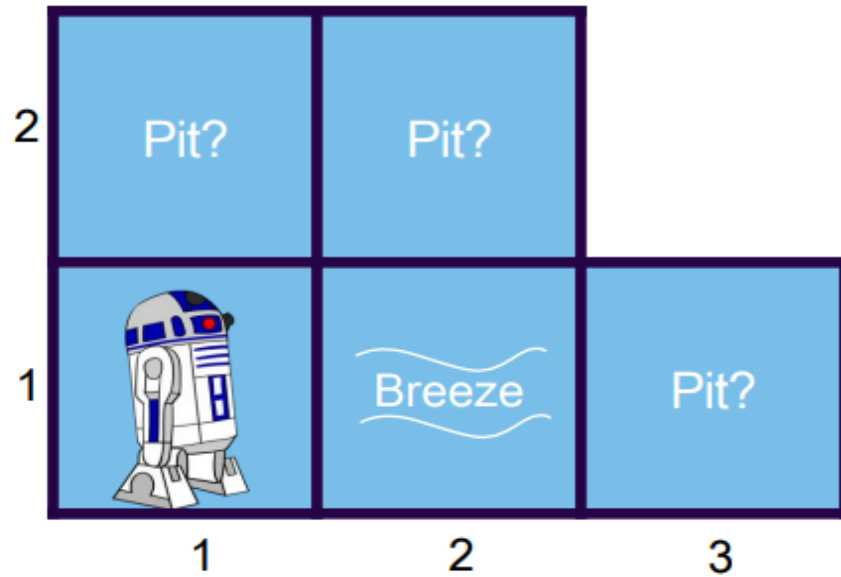
1

2

3

4

Possible Worlds



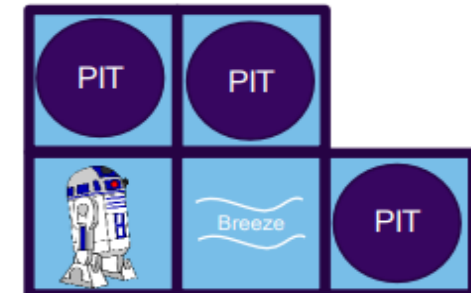
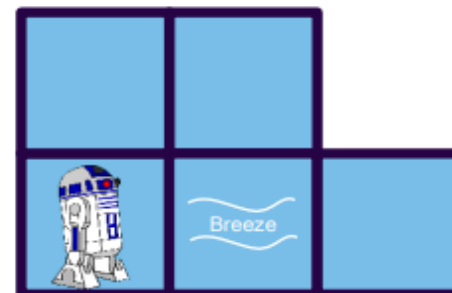
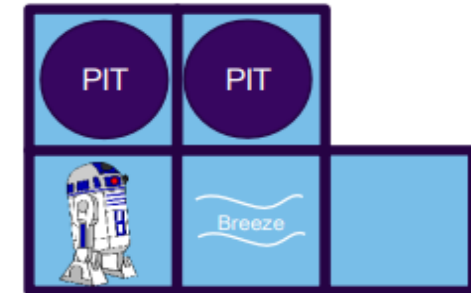
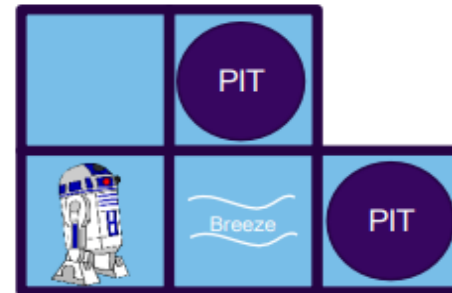
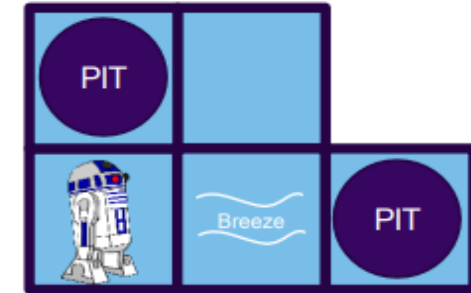
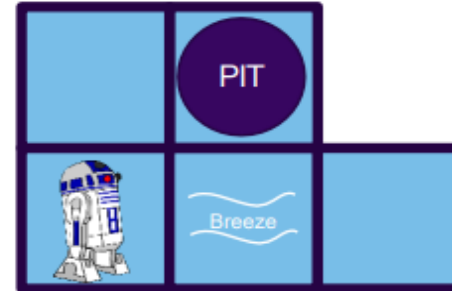
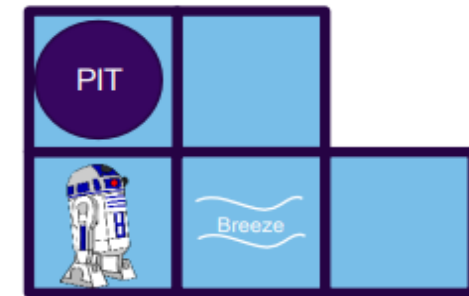
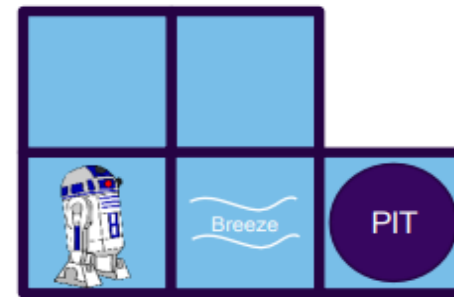
Models are mathematical abstractions that have a fixed set of **truth values** which are **{true, false}** for each sentence.

If sentence α is true in model m then we say

m satisfies α , or

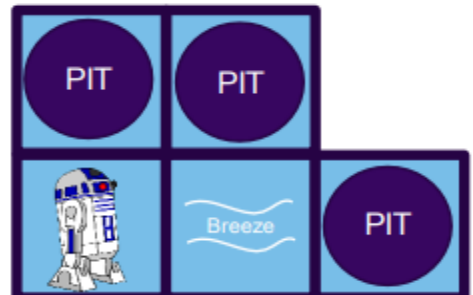
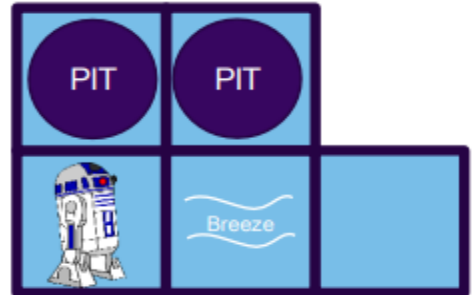
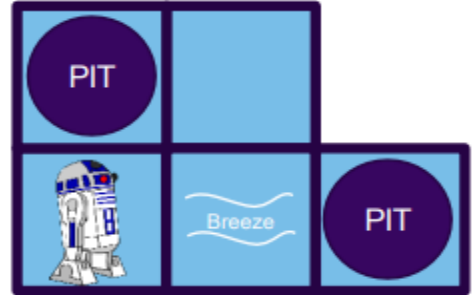
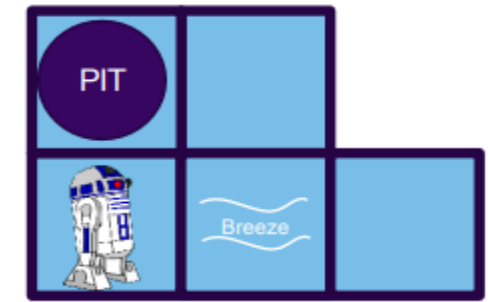
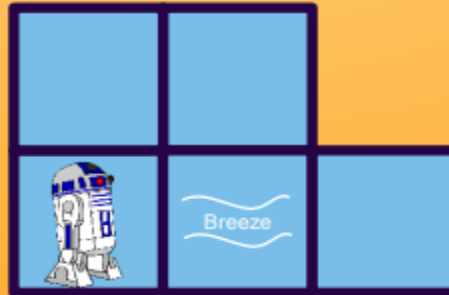
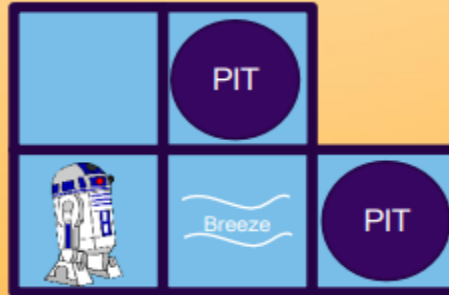
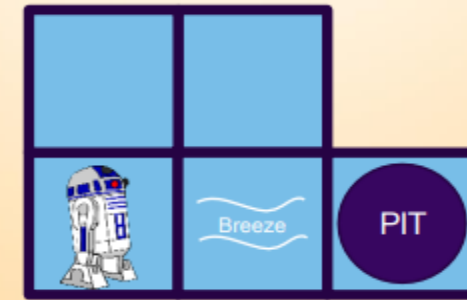
m is a model of α

We use the notation $M(\alpha)$ to mean the **set of all models** of α .



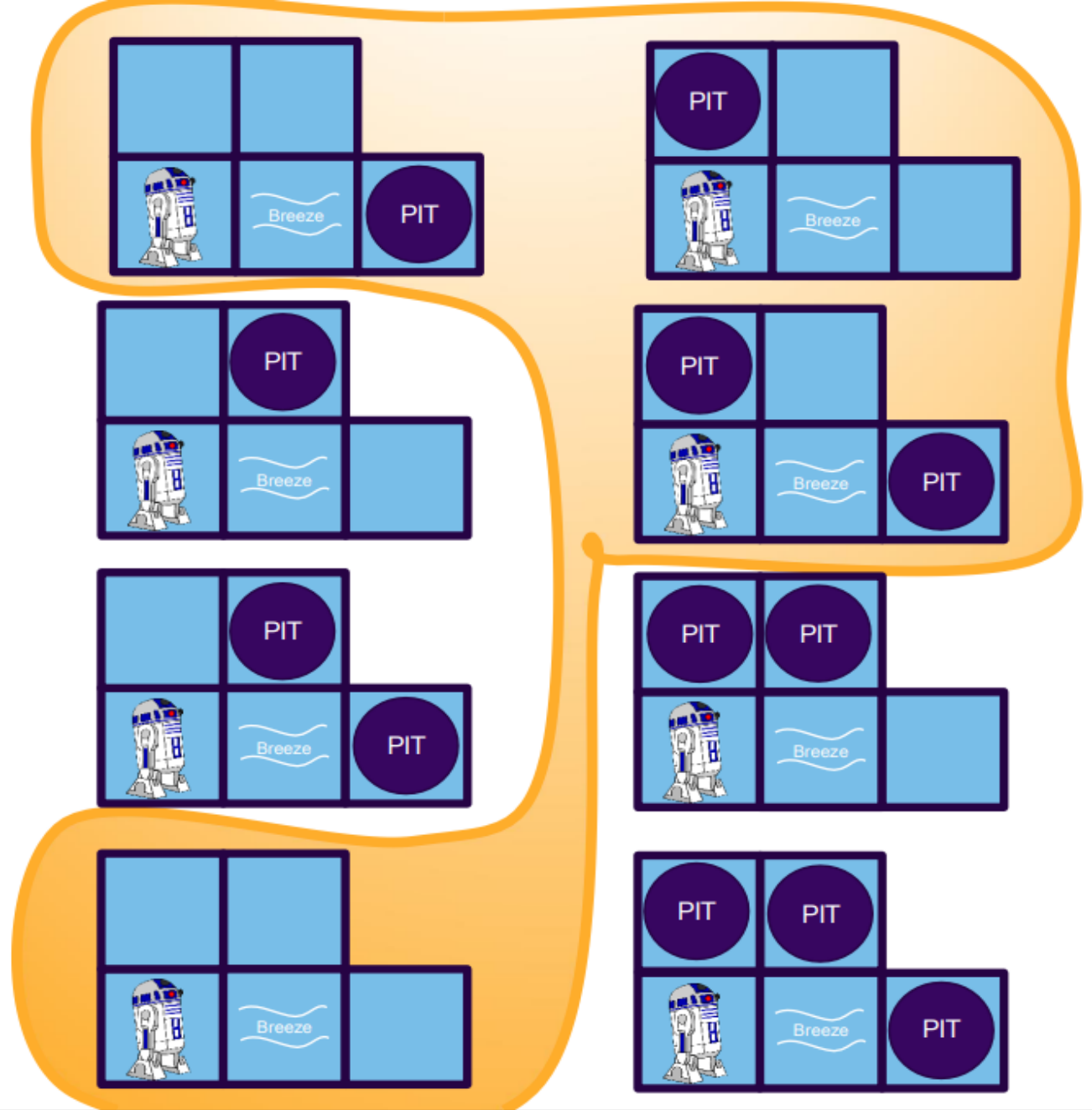
Possible Worlds

α_1 = "There is no pit in [1,2]"



Possible Worlds

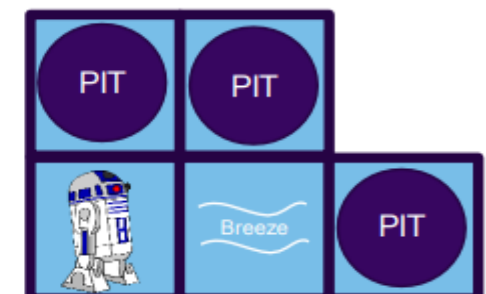
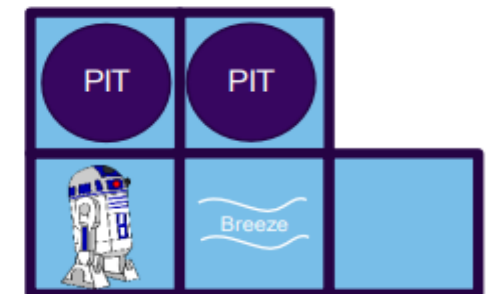
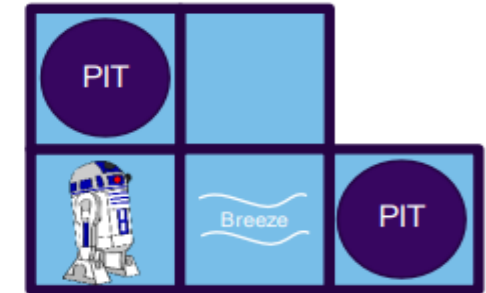
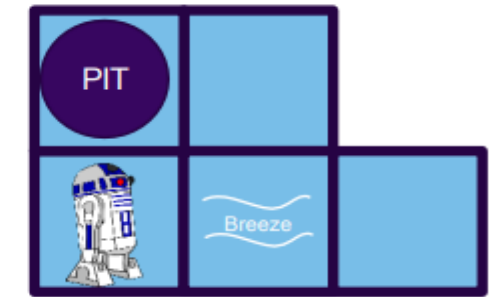
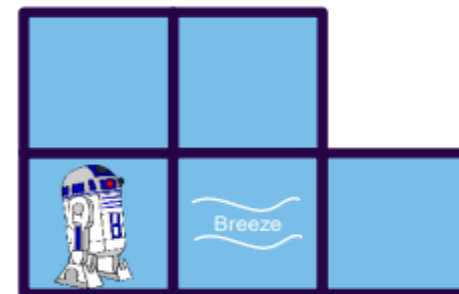
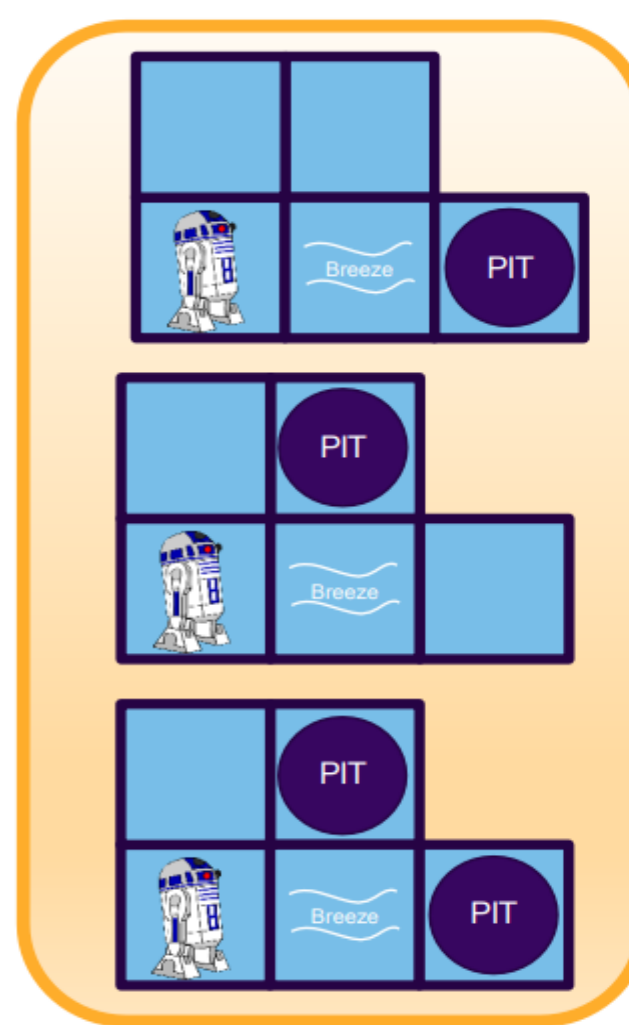
α_2 = "There is no pit in [2,2]"



Possible Worlds

KB =

R1: $\neg P_{1,1}$
 R2: $B_{1,1} \Leftrightarrow (P_{1,2} \vee P_{2,1})$
 R3: $B_{2,1} \Leftrightarrow (P_{1,1} \vee P_{2,2} \vee P_{3,1})$
 R4: $\neg B_{1,1}$
 R5: $B_{2,1}$

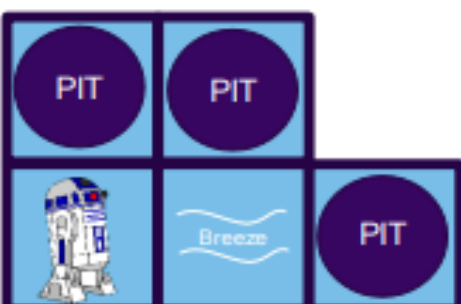
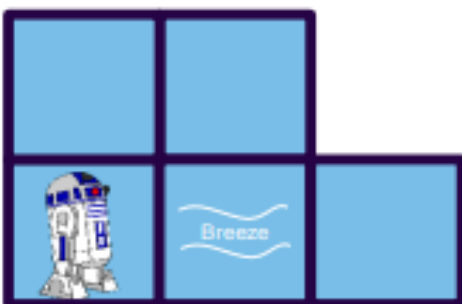
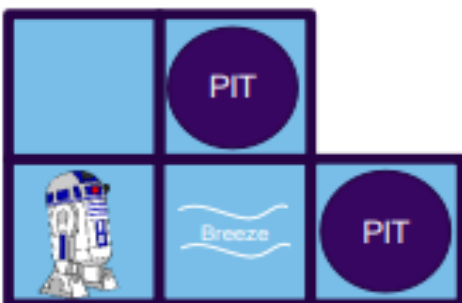
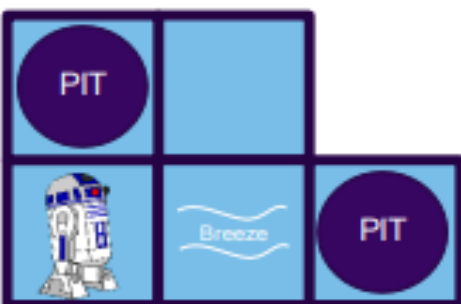
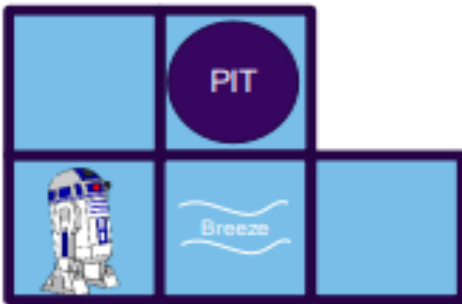
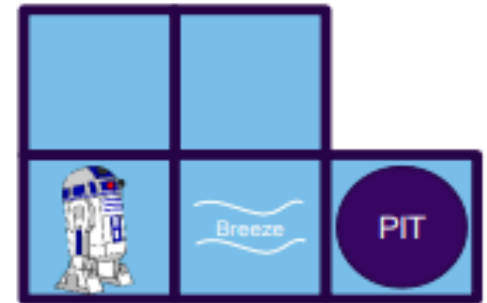


Possible Worlds

$\beta \models \alpha$ if and only if $M(\beta) \subseteq M(\alpha)$

“ β entails α if and only if every model in which β is true, α is also true”

Does our KB entail that there is no pit in [1,2]?
 $KB \models \alpha_1$ if and only if $M(KB) \subseteq M(\alpha_1)$



Possible Worlds

$\beta \models \alpha$ if and only if $M(\beta) \subseteq M(\alpha)$

" β entails α if and only if every model in which β is true, α is also true"

Does our KB entail that there is no pit in [1,2]?

$KB \models \alpha_1$ if and only if $M(KB) \subseteq M(\alpha_1)$

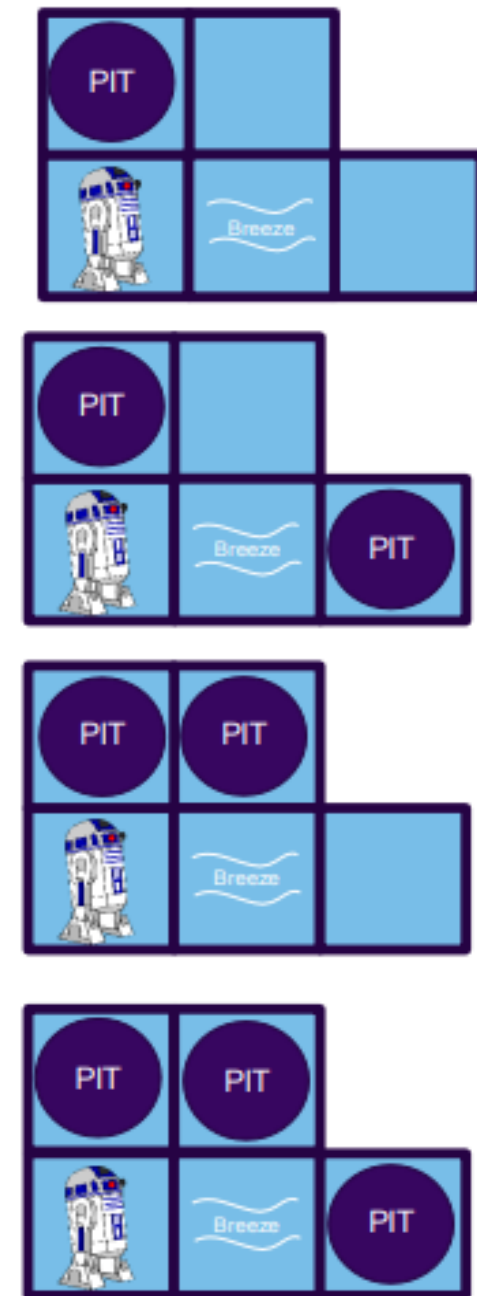
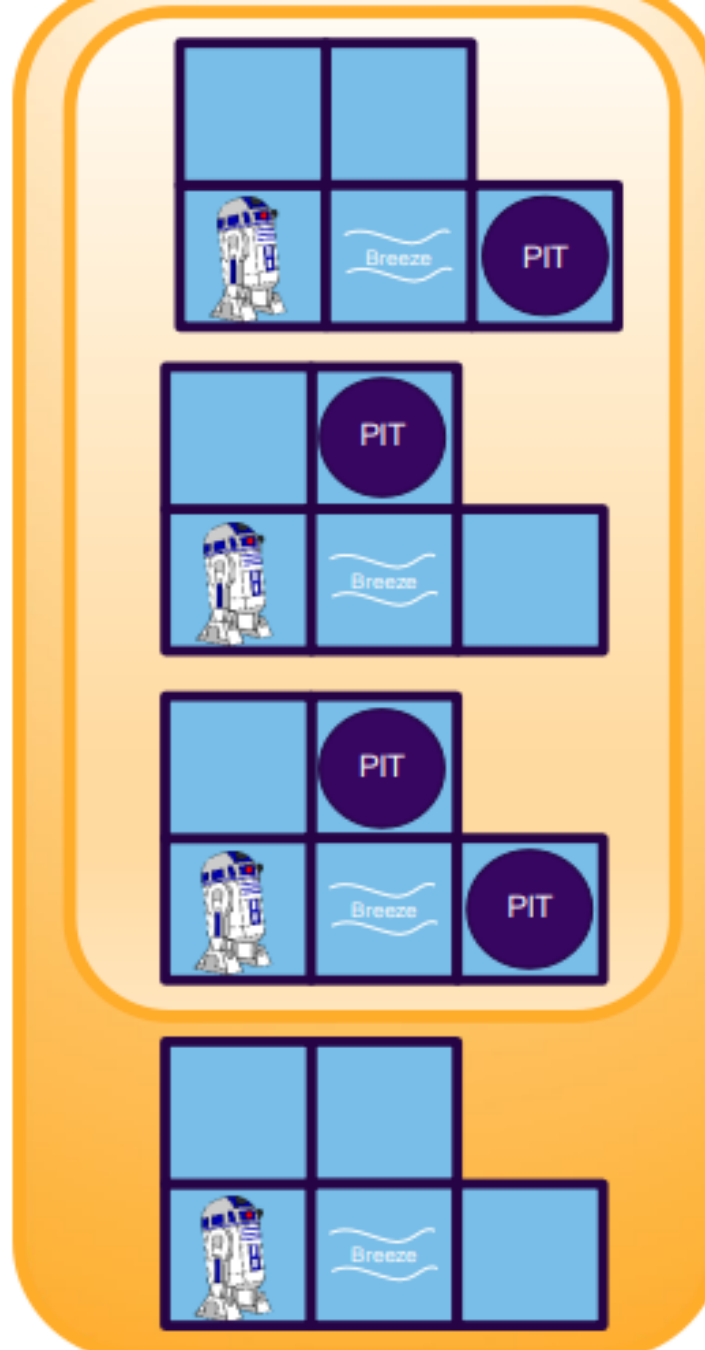
KB =

R1: $\neg P_{1,1}$
 R2: $B_{1,1} \Leftrightarrow (P_{1,2} \vee P_{2,1})$
 R3: $B_{2,1} \Leftrightarrow (P_{1,1} \vee P_{2,2} \vee P_{3,1})$
 R4: $\neg B_{1,1}$
 R5: $B_{2,1}$

α_1 = "There is no pit in [1,2]"

KB $\models \alpha$

1



Possible Worlds

$\beta \models \alpha$ if and only if $M(\beta) \subseteq M(\alpha)$

" β entails α if and only if every model in which β is true, α is also true"

Does our KB entail that there is
no pit in [2,2]?

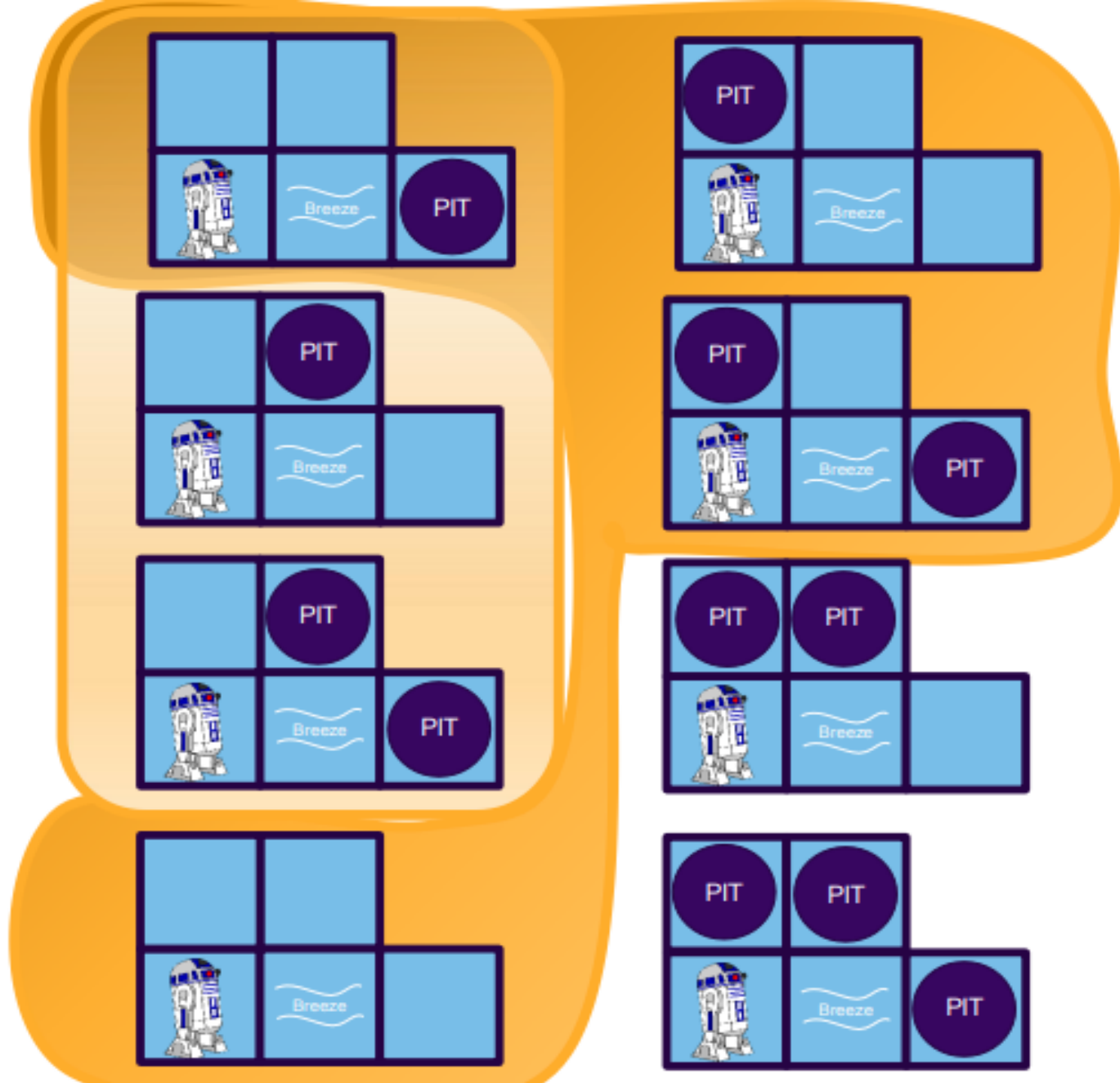
$KB \models \alpha_2$ if and only if $M(KB) \subseteq M(\alpha_2)$

KB =

- R1: $\neg P_{1,1}$
- R2: $B_{1,1} \Leftrightarrow (P_{1,2} \vee P_{2,1})$
- R3: $B_{2,1} \Leftrightarrow (P_{1,1} \vee P_{2,2} \vee P_{3,1})$
- R4: $\neg B_{1,1}$
- R5: $B_{2,1}$

α_2 = "There is no pit in [2,2]"

**KB does not entail α_2 in some models
where KB is true α_2 is false**



Entailment and Inference

Entailment can be applied to derive conclusions, which is the process of **logical inference**.

We can think about the consequences of a KB as a large set of additional sentences that are entailed given the sentences that have been added to the KB.

We would like to design **inference algorithms** to enumerate these sentences.

When an inference algorithm i allows us to conclude that α is true, then we write

$$KB \vdash_i \alpha$$

“ α is derived from KB by i ”

Soundness and Completeness

- **Sound** – the inference algorithm should only derive entailed sentences.
- **Complete** – an inference algorithm is complete if it can derive all sentences that are entailed

Logical Equivalence

Two sentences α and β are **logically equivalent** if they are true in the same set of models. We write this as

$$\alpha \equiv \beta$$

An alternate definition of logical equivalence is that two sentences α and β are logically equivalent if and only if they entail each other.

$$\alpha \equiv \beta \text{ if and only if } \alpha \models \beta \text{ and } \beta \models \alpha.$$

$(\alpha \wedge \beta)$	$\equiv$	$(\beta \wedge \alpha)$	Commutativity of $\wedge$
$(\alpha \vee \beta)$	$\equiv$	$(\beta \vee \alpha)$	Commutativity of $\vee$
$((\alpha \wedge \beta) \wedge \gamma)$	$\equiv$	$(\alpha \wedge (\beta \wedge \gamma))$	Associativity of $\wedge$
$((\alpha \vee \beta) \vee \gamma)$	$\equiv$	$(\alpha \vee (\beta \vee \gamma))$	Associativity of $\vee$
$\neg(\neg \alpha)$	$\equiv$	α	Double-negation elimination
$(\alpha \Rightarrow \beta)$	$\equiv$	$(\neg \beta \Rightarrow \neg \alpha)$	Contraposition
$(\alpha \Rightarrow \beta)$	$\equiv$	$(\neg \alpha \vee \beta)$	Implication elimination
$(\alpha \Leftrightarrow \beta)$	$\equiv$	$((\alpha \Rightarrow \beta) \wedge (\beta \Rightarrow \alpha))$	Biconditional elimination
$\neg(\alpha \wedge \beta)$	$\equiv$	$(\neg \alpha \vee \neg \beta)$	De Morgan
$\neg(\alpha \vee \beta)$	$\equiv$	$(\neg \alpha \wedge \neg \beta)$	De Morgan
$((\alpha \wedge \beta) \vee \gamma)$	$\equiv$	$((\alpha \wedge \beta) \vee (\alpha \wedge \gamma))$	Distributivity of $\wedge$ over $\vee$
$((\alpha \vee \beta) \wedge \gamma)$	$\equiv$	$((\alpha \vee \beta) \wedge (\alpha \vee \gamma))$	Distributivity of $\vee$ over $\wedge$

Inference Rules

Inference Rules can be used to derive proofs. Here's the notation:

Whenever sentences of the form $\alpha \Rightarrow \beta$ and α are given...

$$\frac{\alpha \Rightarrow \beta, \alpha}{\beta}$$

...then the sentence β can be inferred.

Inference Rules

Modus Ponens:

$$\frac{\alpha \Rightarrow \beta, \alpha}{\beta}$$

And Elimination:

$$\frac{\alpha \wedge \beta}{\alpha}$$

Biconditional Elimination:

$$\frac{\alpha \Leftrightarrow \beta}{(\alpha \Rightarrow \beta) \wedge (\beta \Rightarrow \alpha)}$$

Biconditional Elimination:

$$\frac{(\alpha \Rightarrow \beta) \wedge (\beta \Rightarrow \alpha)}{\alpha \Leftrightarrow \beta}$$

All of the logical equivalence rules can be re-written as inference rules.

Inference Example

Our KB

R1: $\neg P_{1,1}$
R2: $B_{1,1} \Leftrightarrow (P_{1,2} \vee P_{2,1})$
R3: $B_{2,1} \Leftrightarrow (P_{1,1} \vee P_{2,2} \vee P_{3,1})$
R4: $\neg B_{1,1}$
R5: $B_{2,1}$

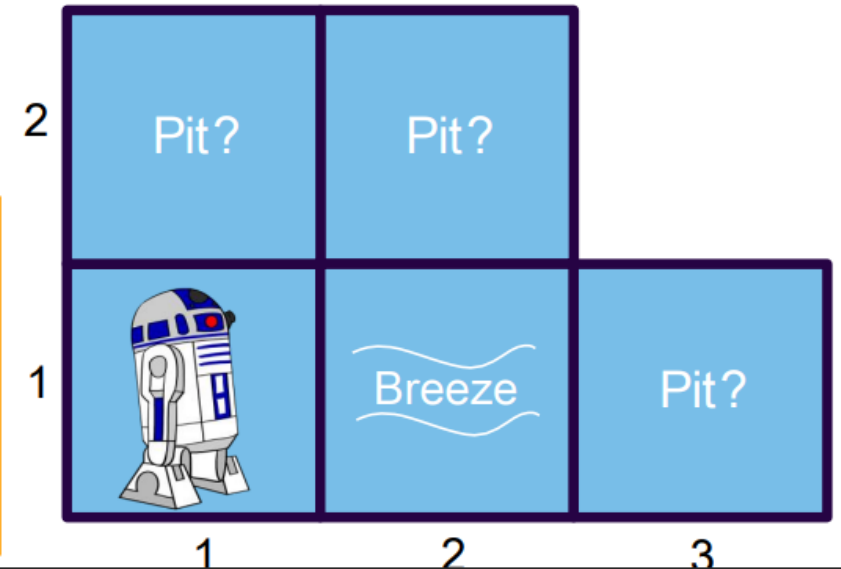
Biconditional Elimination:

$$\frac{\alpha \Leftrightarrow \beta}{(\alpha \Rightarrow \beta) \wedge (\beta \Rightarrow \alpha)}$$

Apply biconditional Elimination to R2 to get R6.

R6: $(B_{1,1} \Rightarrow (P_{1,2} \vee P_{2,1})) \wedge ((P_{1,2} \vee P_{2,1}) \Rightarrow B_{1,1})$

Monotonicity: if $KB \models \alpha$ then $KB \wedge \beta \models \alpha$
We can safely add to the KB, without invalidating anything else that we inferred.



Inference Example

Our KB

R1: $\neg P_{1,1}$

R2: $B_{1,1} \Leftrightarrow (P_{1,2} \vee P_{2,1})$

R3: $B_{2,1} \Leftrightarrow (P_{1,1} \vee P_{2,2} \vee P_{3,1})$

R4: $\neg B_{1,1}$

R5: $B_{2,1}$

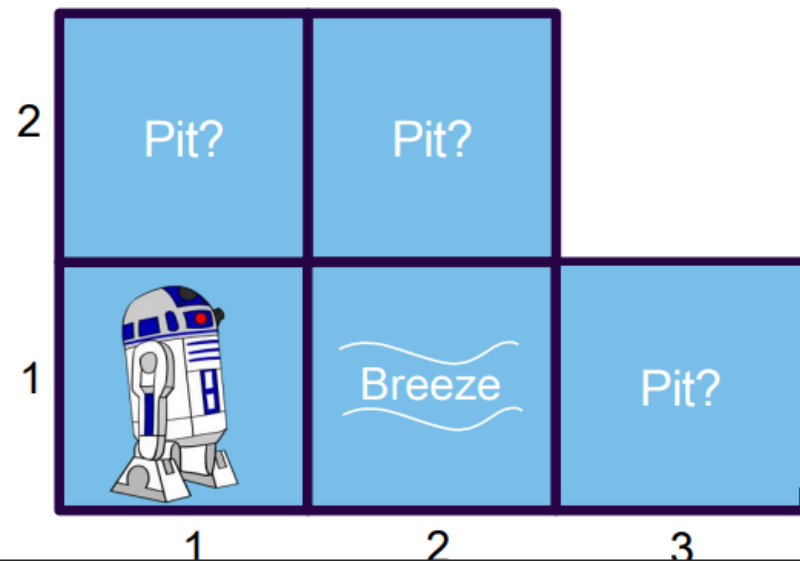
R6: $(B_{1,1} \Rightarrow (P_{1,2} \vee P_{2,1})) \wedge ((P_{1,2} \vee P_{2,1}) \Rightarrow B_{1,1})$

Apply And-Elimination to R6 to get R7.

R7: $(B_{1,1} \Rightarrow (P_{1,2} \vee P_{2,1}))$

And Elimination:

$$\frac{\alpha \wedge \beta}{\alpha}$$



Inference Example

Our KB

R1: $\neg P_{1,1}$

R2: $B_{1,1} \Leftrightarrow (P_{1,2} \vee P_{2,1})$

R3: $B_{2,1} \Leftrightarrow (P_{1,1} \vee P_{2,2} \vee P_{3,1})$

R4: $\neg B_{1,1}$

R5: $B_{2,1}$

R6: $(B_{1,1} \Rightarrow (P_{1,2} \vee P_{2,1})) \wedge ((P_{1,2} \vee P_{2,1}) \Rightarrow B_{1,1})$

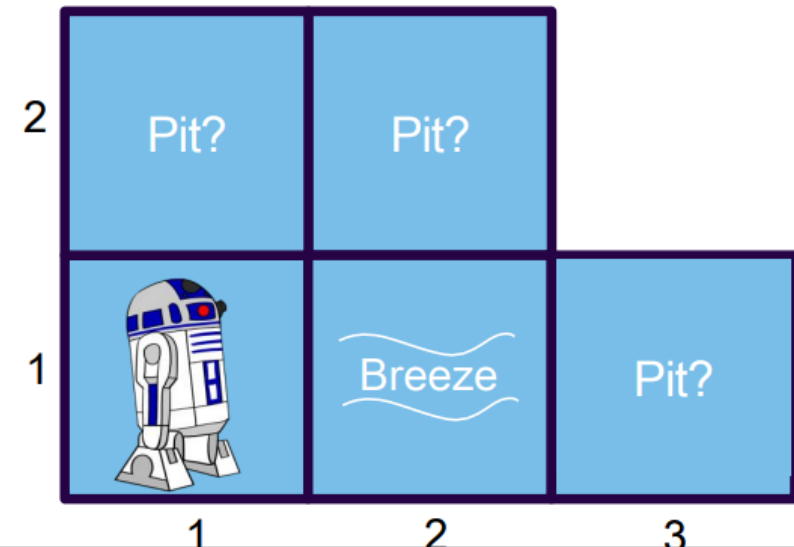
R7: $((P_{1,2} \vee P_{2,1}) \Rightarrow B_{1,1})$

Logical equivalence for contrapositives applied to R7 gives R8.

R8: $(\neg B_{1,1} \Rightarrow \neg (P_{1,2} \vee P_{2,1}))$

Logical equivalence for contrapositives:

$$\frac{(\alpha \Rightarrow \beta)}{(\neg \beta \Rightarrow \neg \alpha)}$$



Inference Example

Our KB

R1: $\neg P_{1,1}$
R2: $B_{1,1} \Leftrightarrow (P_{1,2} \vee P_{2,1})$
R3: $B_{2,1} \Leftrightarrow (P_{1,1} \vee P_{2,2} \vee P_{3,1})$
R4: $\neg B_{1,1}$
R5: $B_{2,1}$

R6: $(B_{1,1} \Rightarrow (P_{1,2} \vee P_{2,1})) \wedge ((P_{1,2} \vee P_{2,1}) \Rightarrow B_{1,1})$

R7: $((P_{1,2} \vee P_{2,1}) \Rightarrow B_{1,1})$

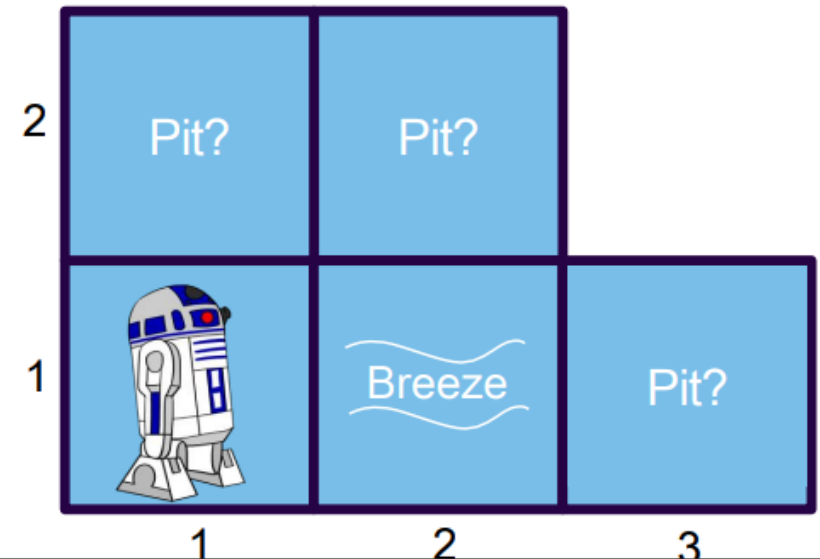
R8: $(\neg B_{1,1} \Rightarrow \neg (P_{1,2} \vee P_{2,1}))$

Apply Modus Ponens to R4 and R8 to get:

R9: $\neg (P_{1,2} \vee P_{2,1})$

Modus Ponens:

$$\frac{\alpha \Rightarrow \beta, \alpha}{\beta}$$



Inference Example

Our KB

R1: $\neg P_{1,1}$
R2: $B_{1,1} \Leftrightarrow (P_{1,2} \vee P_{2,1})$
R3: $B_{2,1} \Leftrightarrow (P_{1,1} \vee P_{2,2} \vee P_{3,1})$
R4: $\neg B_{1,1}$
R5: $B_{2,1}$

R6: $(B_{1,1} \Rightarrow (P_{1,2} \vee P_{2,1})) \wedge ((P_{1,2} \vee P_{2,1}) \Rightarrow B_{1,1})$

R7: $((P_{1,2} \vee P_{2,1}) \Rightarrow B_{1,1})$

R8: $(\neg B_{1,1} \Rightarrow \neg (P_{1,2} \vee P_{2,1}))$

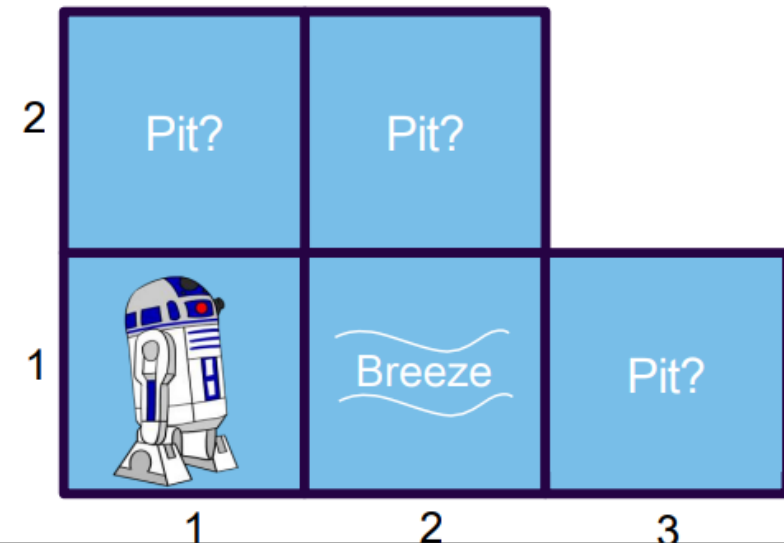
R9: $\neg (P_{1,2} \vee P_{2,1})$

Apply De Morgan's Rule to R9:

R10: $\neg P_{1,2} \wedge \neg P_{2,1}$

De Morgan's Rule

$$\frac{\neg(\alpha \vee \beta)}{(\neg\alpha \wedge \neg\beta)}$$



Inference Example

Our KB

R1: $\neg P_{1,1}$
R2: $B_{1,1} \Leftrightarrow (P_{1,2} \vee P_{2,1})$
R3: $B_{2,1} \Leftrightarrow (P_{1,1} \vee P_{2,2} \vee P_{3,1})$
R4: $\neg B_{1,1}$
R5: $B_{2,1}$

R6: $(B_{1,1} \Rightarrow (P_{1,2} \vee P_{2,1})) \wedge ((P_{1,2} \vee P_{2,1}) \Rightarrow B_{1,1})$

R7: $((P_{1,2} \vee P_{2,1}) \Rightarrow B_{1,1})$

R8: $(\neg B_{1,1} \Rightarrow \neg (P_{1,2} \vee P_{2,1}))$

R9: $\neg (P_{1,2} \vee P_{2,1})$

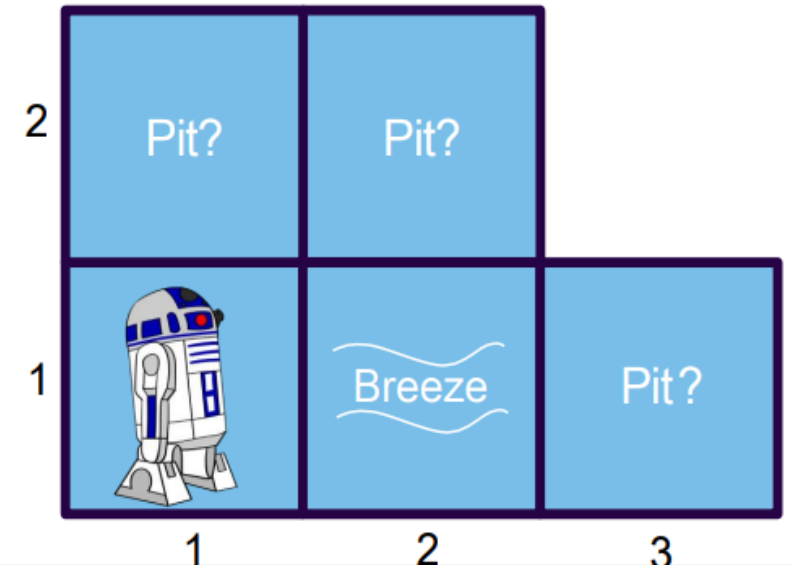
R10: $\neg P_{1,2} \wedge \neg P_{2,1}$

R11: $\neg P_{1,2}$

R12: $\neg P_{2,1}$

And Elimination:

$$\frac{\alpha \wedge \beta}{\alpha}$$



Inference Example

Our KB

R1: $\neg P_{1,1}$

R2: $B_{1,1} \Leftrightarrow (P_{1,2} \vee P_{2,1})$

R3: $B_{2,1} \Leftrightarrow (P_{1,1} \vee P_{2,2} \vee P_{3,1})$

R4: $\neg B_{1,1}$

R5: $B_{2,1}$

R6: $(B_{1,1} \Rightarrow (P_{1,2} \vee P_{2,1})) \wedge ((P_{1,2} \vee P_{2,1}) \Rightarrow B_{1,1})$

R7: $((P_{1,2} \vee P_{2,1}) \Rightarrow B_{1,1})$

R8: $(\neg B_{1,1} \Rightarrow \neg (P_{1,2} \vee P_{2,1}))$

R9: $\neg (P_{1,2} \vee P_{2,1})$

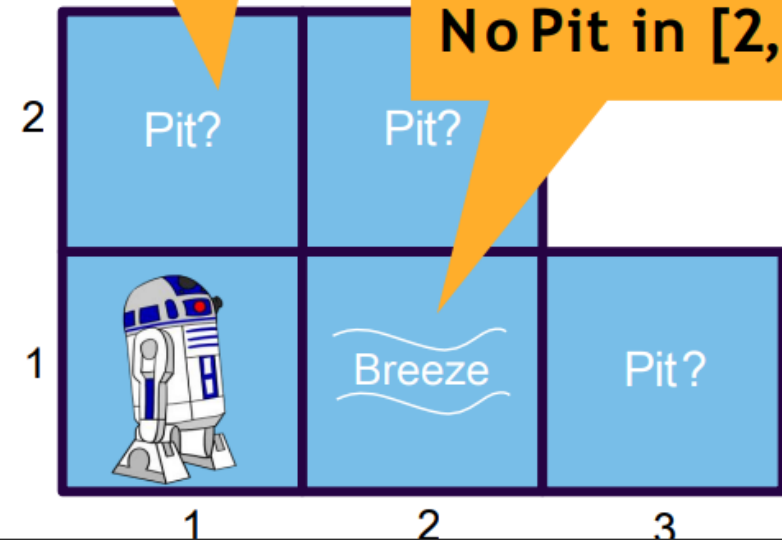
R10: $\neg P_{1,2} \wedge \neg P_{2,1}$

R11: $\neg P_{1,2}$

R12: $\neg P_{2,1}$

No Pit in [1,2].

No Pit in [2,1].



Search for a Proof

We can use search algorithms to find a sequence of steps that constitutes a proof!

- **Initial State:** the initial knowledge base.
- **Actions:** All inference rules applied to all sentence that match the top half of the inference rule.
- **Result:** Add the sentence on the bottom half to the KB.
- **Goal:** The goal is a state that contains the sentence that we are trying to prove.

Initial State

R1: $\neg P_{1,1}$
R2: $B_{1,1} \Leftrightarrow (P_{1,2} \vee P_{2,1})$
R3: $B_{2,1} \Leftrightarrow (P_{1,1} \vee P_{2,2} \vee P_{3,1})$
R4: $\neg B_{1,1}$
R5: $B_{2,1}$

Action

$B_{1,1} \Leftrightarrow (P_{1,2} \vee P_{2,1})$

$(B_{1,1} \Rightarrow (P_{1,2} \vee P_{2,1})) \wedge ((P_{1,2} \vee P_{2,1}) \Rightarrow B_{1,1})$

Result

R1: $\neg P_{1,1}$
R2: $B_{1,1} \Leftrightarrow (P_{1,2} \vee P_{2,1})$
R3: $B_{2,1} \Leftrightarrow (P_{1,1} \vee P_{2,2} \vee P_{3,1})$
R4: $\neg B_{1,1}$
R6: $B_{2,1}$
R7: $(B_{1,1} \Rightarrow (P_{1,2} \vee P_{2,1})) \wedge ((P_{1,2} \vee P_{2,1}) \Rightarrow B_{1,1})$

Goal

$\neg P_{1,2}$

Monotonicity implies Soundness

Monotonicity says that the set of entailed sentence can only increase as information is added to the KB.

if $KB \models \alpha$ then $KB \wedge \beta \models \alpha$

If we add additional information to the KB, it doesn't invalidate any other info that we've already inferred.

Therefore it is **sound** it will only derive entailed sentences.

Proof by Resolution

But is it **complete**? Can this method derive all sentences that are entailed? If we used iterative deepening search, then we would find any reachable goal. But what if we were missing an important inference rule?

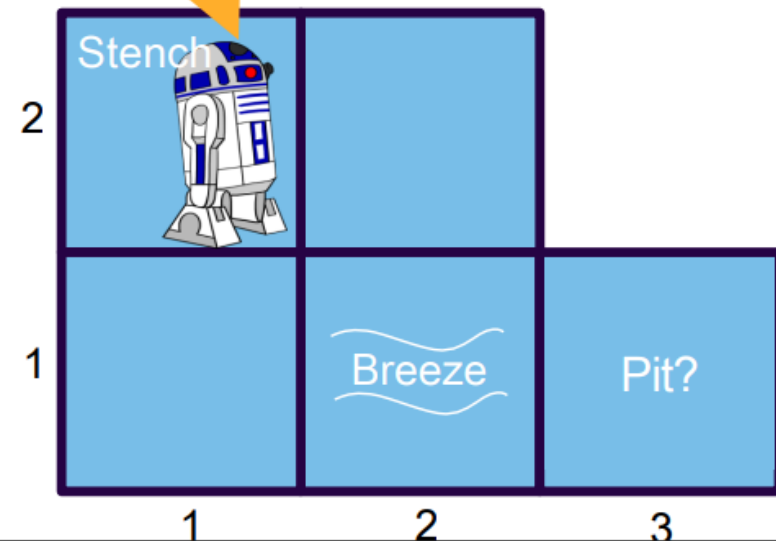
There is a single inference rule, **resolution**, which yields a complete inference algorithm when coupled with any complete search algorithm.

R1: $\neg P_{1,1}$
 R2: $B_{1,1} \Leftrightarrow (P_{1,2} \vee P_{2,1})$
 R3: $B_{2,1} \Leftrightarrow (P_{1,1} \vee P_{2,2} \vee P_{3,1})$
 R4: $\neg B_{1,1}$
 R5: $B_{2,1}$
 R6: $(B_{1,1} \Rightarrow (P_{1,2} \vee P_{2,1})) \wedge ((P_{1,2} \vee P_{2,1}) \Rightarrow B_{1,1})$
 R7: $((P_{1,2} \vee P_{2,1}) \Rightarrow B_{1,1})$
 R8: $(\neg B_{1,1} \Rightarrow \neg (P_{1,2} \vee P_{2,1}))$
 R9: $\neg (P_{1,2} \vee P_{2,1})$
 R10: $\neg P_{1,2} \wedge \neg P_{2,1}$
 R11: $\neg P_{1,2}$
 R12: $\neg P_{2,1}$

R13: $\neg B_{2,1}$
 R14: $B_{1,2} \Leftrightarrow (P_{1,1} \vee P_{2,2} \vee P_{1,3})$
 R15: $\neg P_{2,2}$
 R16: $\neg P_{1,3}$
 R17: $P_{1,1} \vee P_{2,2} \vee P_{3,1}$

New Rules

Stench, but no Breeze



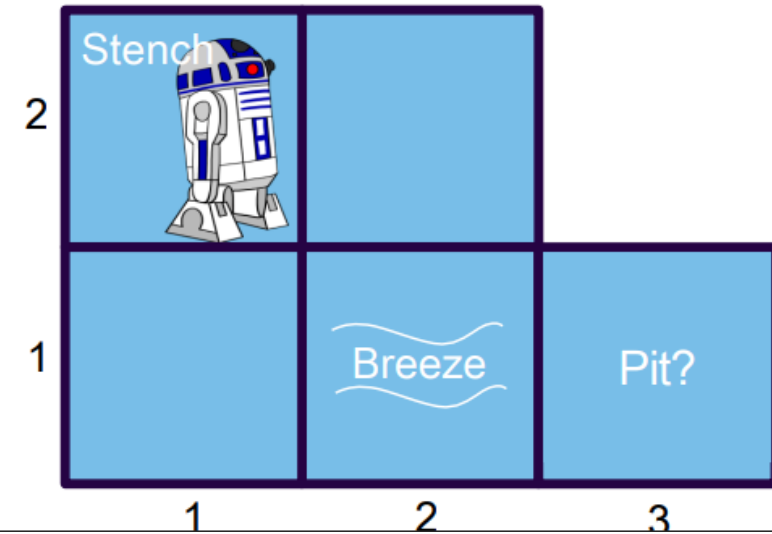
R1: $\neg P_{1,1}$
 R2: $B_{1,1} \Leftrightarrow (P_{1,2} \vee P_{2,1})$
 R3: $B_{2,1} \Leftrightarrow (P_{1,1} \vee P_{2,2} \vee P_{3,1})$
 R4: $\neg B_{1,1}$
 R5: $B_{2,1}$
 R6: $(B_{1,1} \Rightarrow (P_{1,2} \vee P_{2,1})) \wedge ((P_{1,2} \vee P_{2,1}) \Rightarrow B_{1,1})$
 R7: $((P_{1,2} \vee P_{2,1}) \Rightarrow B_{1,1})$
 R8: $(\neg B_{1,1} \Rightarrow \neg (P_{1,2} \vee P_{2,1}))$
 R9: $\neg (P_{1,2} \vee P_{2,1})$
 R10: $\neg P_{1,2} \wedge \neg P_{2,1}$
 R11: $\neg P_{1,2}$
 R12: $\neg P_{2,1}$

R13: $\neg B_{2,1}$
 R14: $B_{1,2} \Leftrightarrow (P_{1,1} \vee P_{2,2} \vee P_{1,3})$
 R15: $\neg P_{2,2}$
 R16: $\neg P_{1,3}$
 R17: $P_{1,1} \vee P_{2,2} \vee P_{3,1}$

R18: $(P_{1,1} \vee P_{3,1})$

First application of **Resolution** using R17 and R15:

$$\frac{(P_{1,1} \vee P_{2,2} \vee P_{3,1}) \wedge \neg P_{2,2}}{(P_{1,1} \vee P_{3,1})}$$



R1: $\neg P_{1,1}$

R2: $B_{1,1} \Leftrightarrow (P_{1,2} \vee P_{2,1})$

R3: $B_{2,1} \Leftrightarrow (P_{1,1} \vee P_{2,2} \vee P_{3,1})$

R4: $\neg B_{1,1}$

R5: $B_{2,1}$

R6: $(B_{1,1} \Rightarrow (P_{1,2} \vee P_{2,1})) \wedge ((P_{1,2} \vee P_{2,1}) \Rightarrow B_{1,1})$

R7: $((P_{1,2} \vee P_{2,1}) \Rightarrow B_{1,1})$

R8: $(\neg B_{1,1} \Rightarrow \neg (P_{1,2} \vee P_{2,1}))$

R9: $\neg (P_{1,2} \vee P_{2,1})$

R10: $\neg P_{1,2} \wedge \neg P_{2,1}$

R11: $\neg P_{1,2}$

R12: $\neg P_{2,1}$

R13: $\neg B_{2,1}$

R14: $B_{1,2} \Leftrightarrow (P_{1,1} \vee P_{2,2} \vee P_{3,3})$

R15: $\neg P_{2,2}$

R16: $\neg P_{1,3}$

R17: $P_{1,1} \vee P_{2,2} \vee P_{3,1}$

R18: $(P_{1,1} \vee P_{3,1})$

R19: $P_{3,1}$

And there's no pit in [1,1]

First application of Resolution using R18 and R1:

$$\frac{(P_{1,1} \vee P_{3,1}) \wedge \neg P_{1,1}}{P_{3,1}}$$

If there is a pit in [1,1]
or a pit in [3,1]

Then there must be a pit in [3,1]

Yes, there is a
Pit in [3,1].

